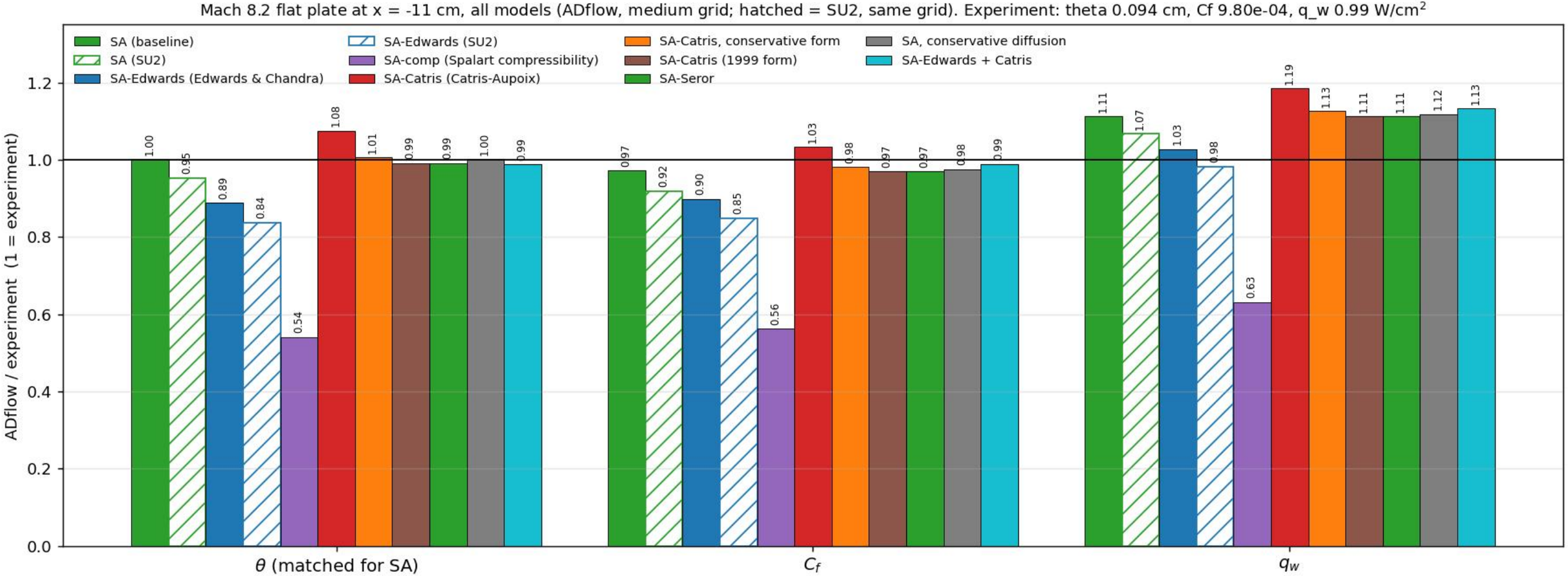
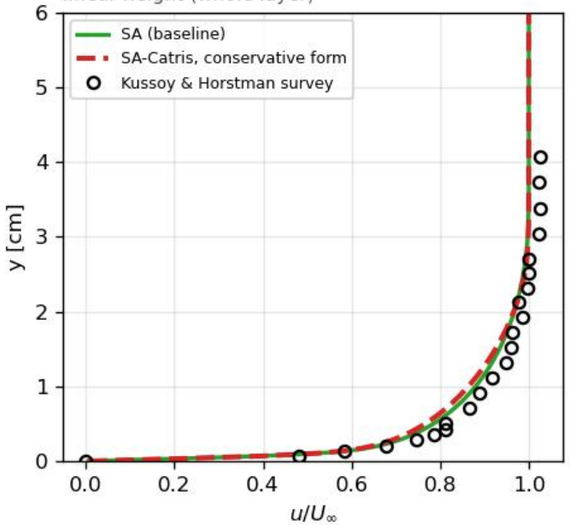
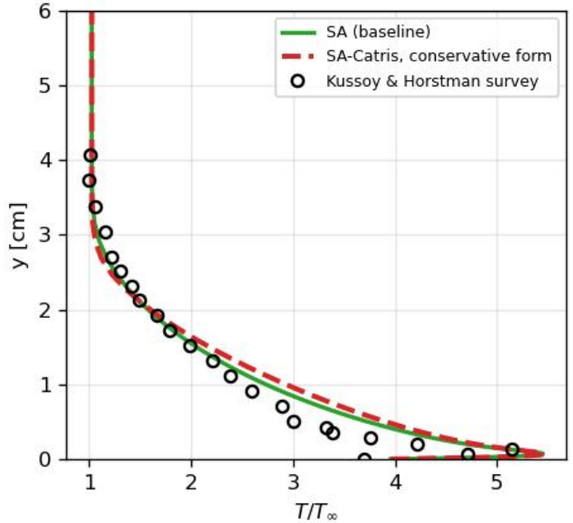
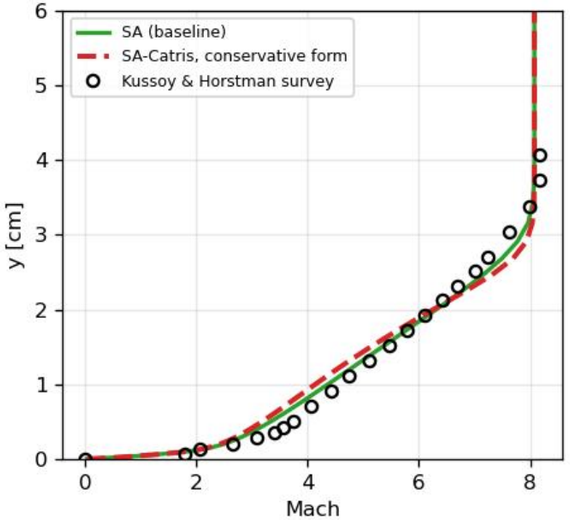

Mach 8.2 flat plate at x = -11 cm, all models (ADflow, medium grid; hatched = SU2, same grid). Experiment: theta 0.094 cm, Cf 9.80e-04, q_w 0.99 W/cm²
SA (baseline)
SA (SU2)
SA-Edwards (Edwards & Chandra)
SA-Edwards (SU2)
SA-comp (Spalart compressibility)
SA-Catris (Catris-Aupoix)
SA-Catris, conservative form
SA-Catris (1999 form)
SA-Seror
SA, conservative diffusion
SA-Edwards + Catris
ADflow / experiment (1 = experiment)
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.00
0.95
0.89
0.84
0.54
1.08
1.01
0.99
0.99
1.00
0.99
0.97
0.92
0.90
0.85
0.56
1.03
0.98
0.97
0.97
0.98
0.99
1.11
1.07
1.03
0.98
0.63
1.19
1.13
1.11
1.11
1.12
1.13
θ (matched for SA)
$C_f$
$q_w$

Mach 8.2 flat plate, incoming boundary layer at x = -11 cm: SA-Catris, conservative form vs SA vs experiment
Catris-Aupoix density correction, conservative left-hand side (TMR SA-noft2-CatrisCons)

linear height (whole layer)



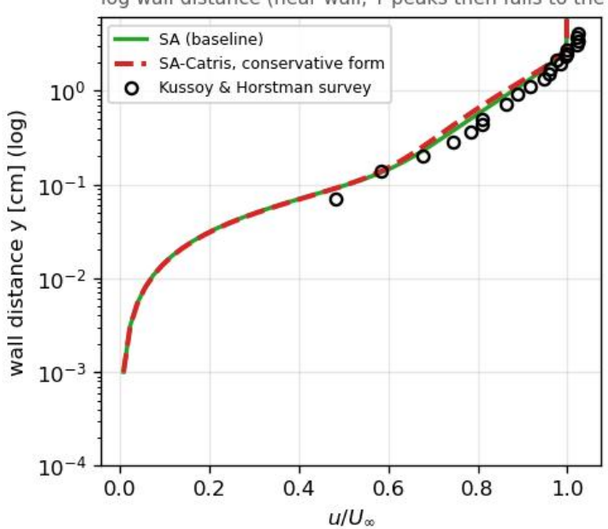



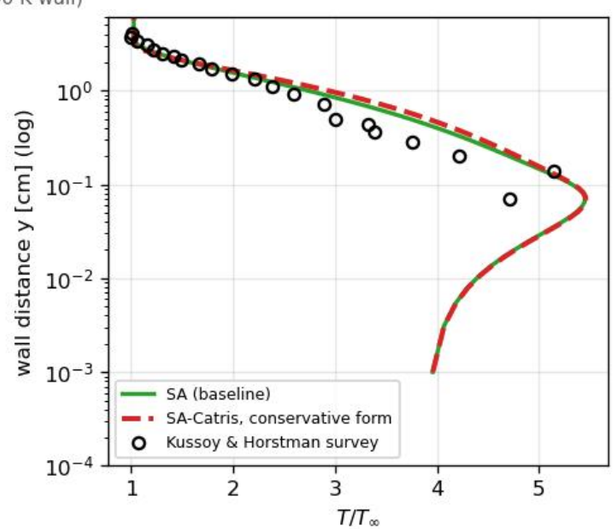



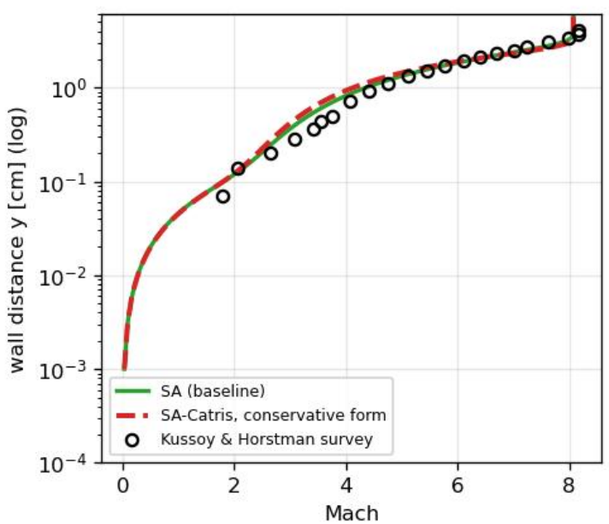

x = -11 cm; model = SA-Catris-cons
(theta matches for SA by construction)

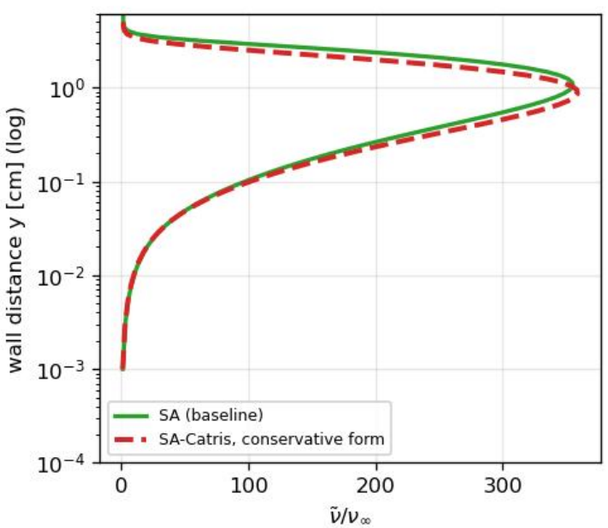
| quantity | exp. | SA | model | model vs SA | model vs exp. |
|---|---|---|---|---|---|
| theta, cm | 0.0940 | 0.0942 | 0.0946 | +0.4% | +0.6% |
| d99, cm | - | 2.635 | 2.523 | -4.3% | - |
| Cf x 1e3 | 0.980 | 0.953 | 0.962 | +1.0% | -1.8% |
| q_w, W/cm2 | 0.990 | 1.102 | 1.116 | +1.2% | +12.7% |

log wall distance (near wall; T peaks then falls to the 300 K wall)



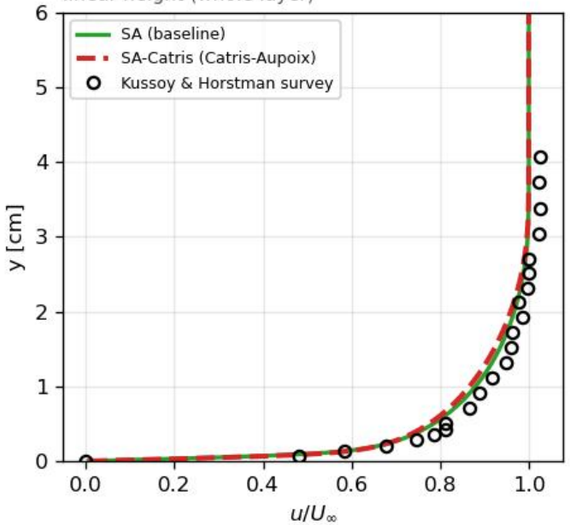



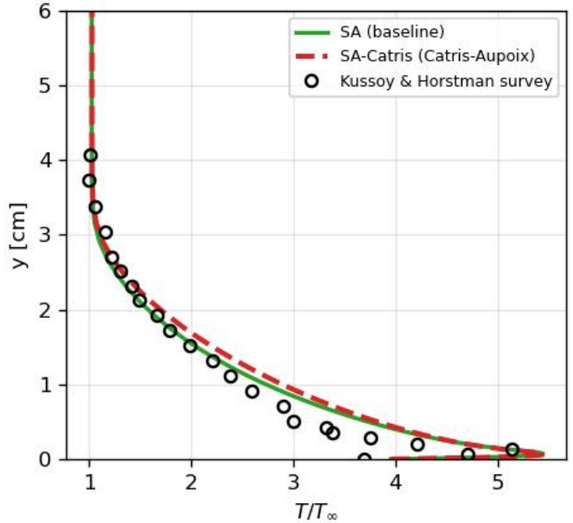



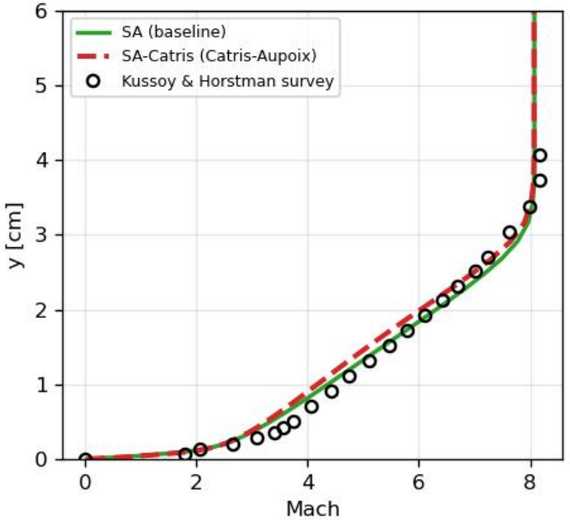



Mach 8.2 flat plate, incoming boundary layer at x = -11 cm:  SA-Catris (Catris-Aupoix)  vs  SA  vs  experiment
Catris-Aupoix density correction (2000): diffuses sqrt(rho) nuTilde, laminar diffusion of mu, original non-conservative left-hand side

linear height (whole layer)



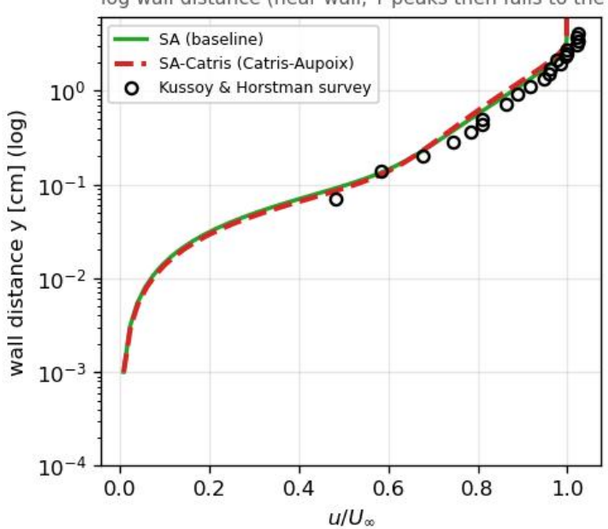



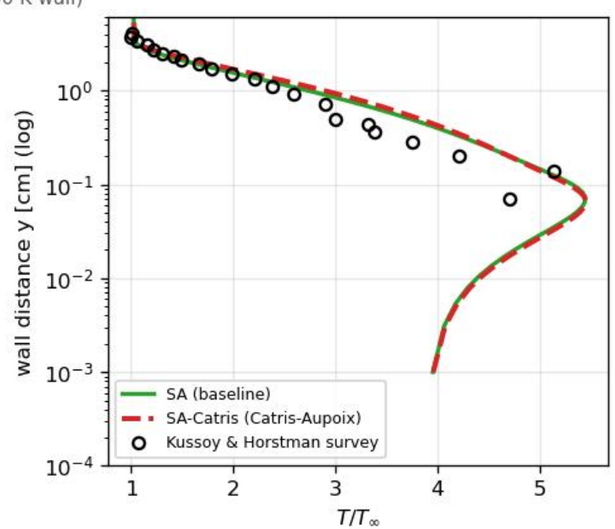



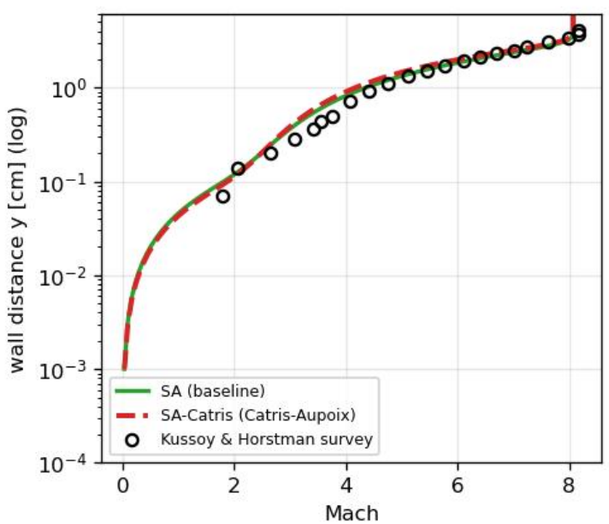

x = -11 cm; model = SA-Catris
(theta matches for SA by construction)

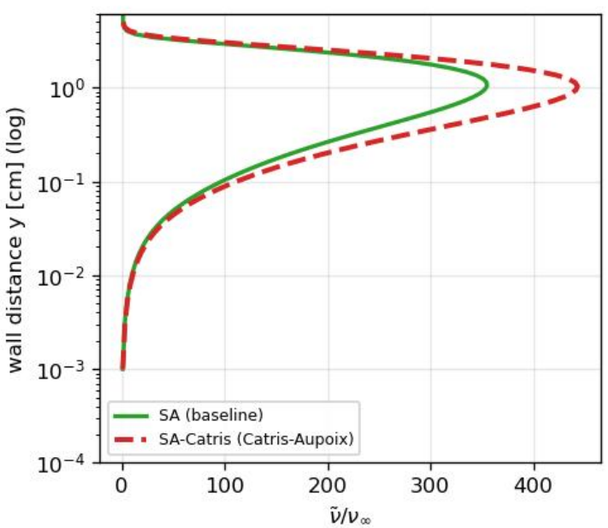
| quantity | exp. | SA | model | model vs SA | model vs exp. |
|---|---|---|---|---|---|
| theta, cm | 0.0940 | 0.0942 | 0.1011 | +7.4% | +7.6% |
| d99, cm | - | 2.635 | 2.757 | +4.6% | - |
| Cf x 1e3 | 0.980 | 0.953 | 1.013 | +6.3% | +3.4% |
| q_w, W/cm2 | 0.990 | 1.102 | 1.173 | +6.5% | +18.5% |

log wall distance (near wall; T peaks then falls to the 300 K wall)



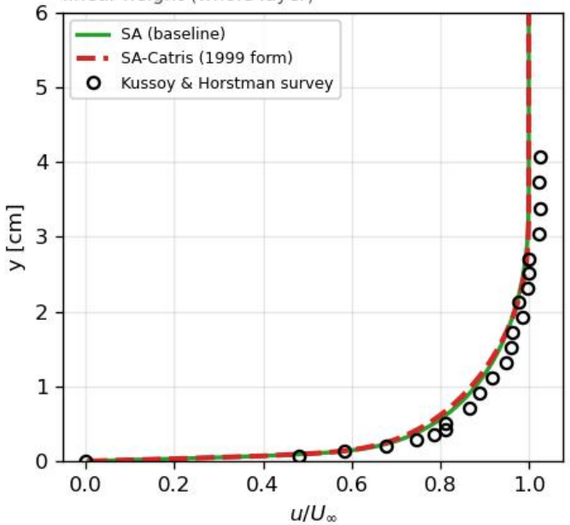



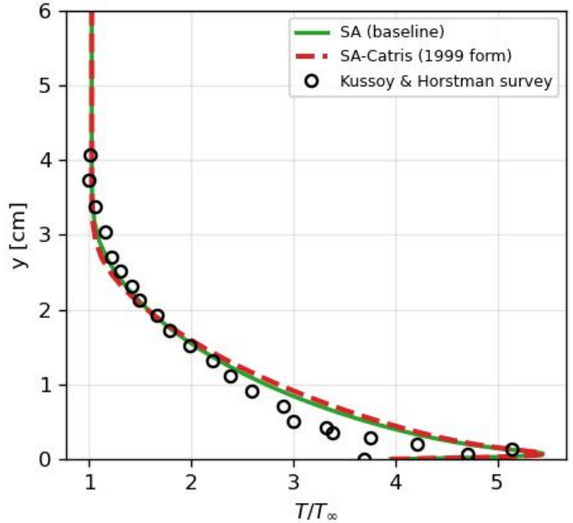



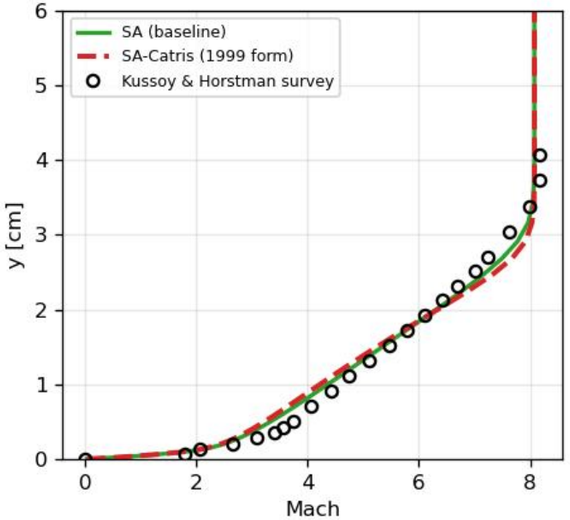



Mach 8.2 flat plate, incoming boundary layer at x = -11 cm: SA-Catris (1999 form) vs SA vs experiment
Catris (1999) form of the density-gradient diffusion terms

linear height (whole layer)



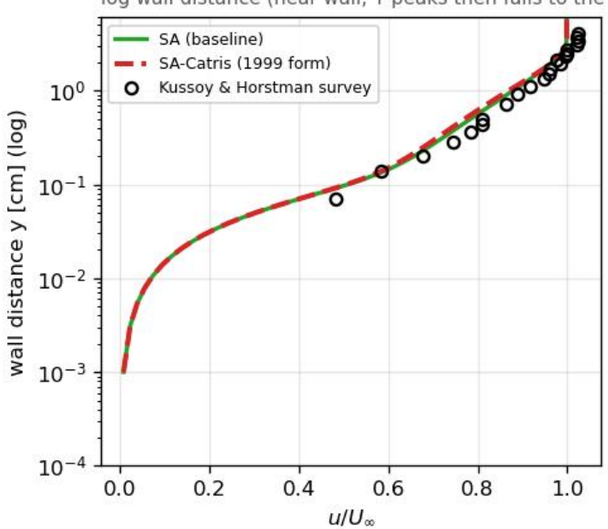



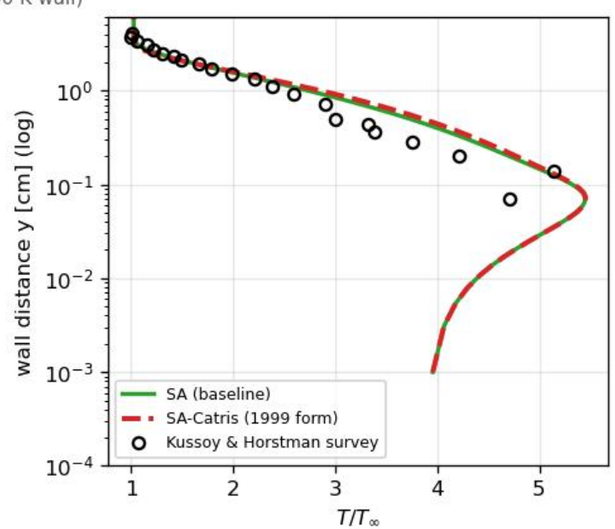



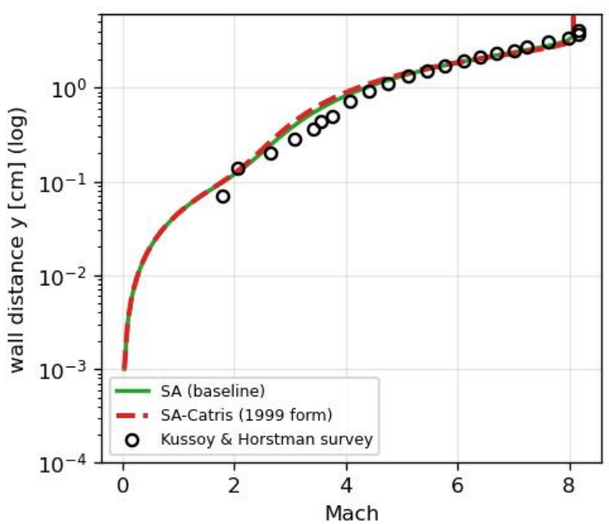

x = -11 cm; model = SA-Catris1999
(theta matches for SA by construction)

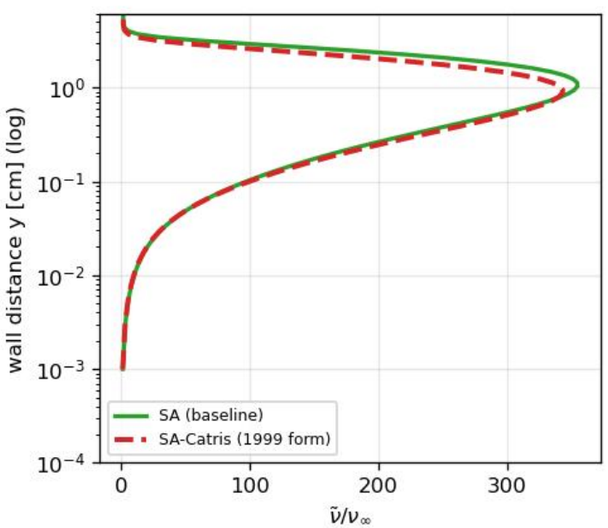
| quantity | exp. | SA | model | model vs SA | model vs exp. |
|---|---|---|---|---|---|
| theta, cm | 0.0940 | 0.0942 | 0.0932 | -1.0% | -0.8% |
| d99, cm | - | 2.635 | 2.526 | -4.2% | - |
| Cf x 1e3 | 0.980 | 0.953 | 0.951 | -0.2% | -3.0% |
| q_w, W/cm2 | 0.990 | 1.102 | 1.102 | -0.0% | +11.3% |

log wall distance (near wall; T peaks then falls to the 300 K wall)



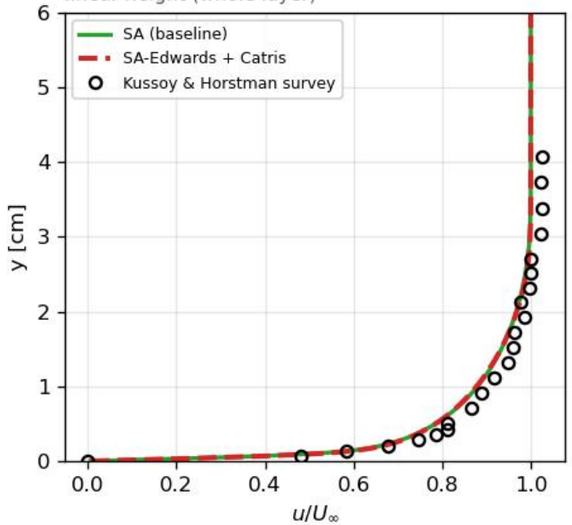



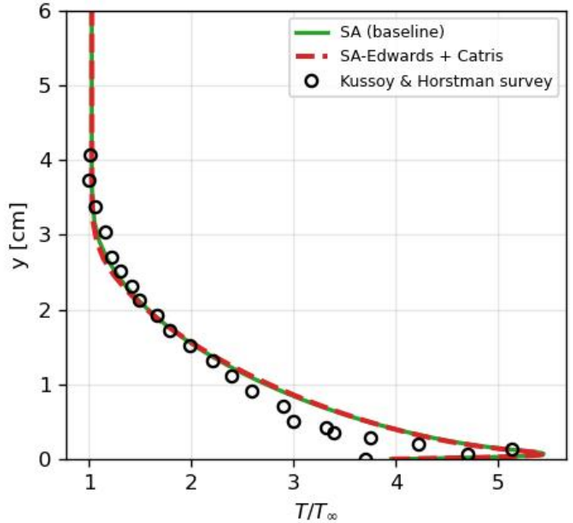



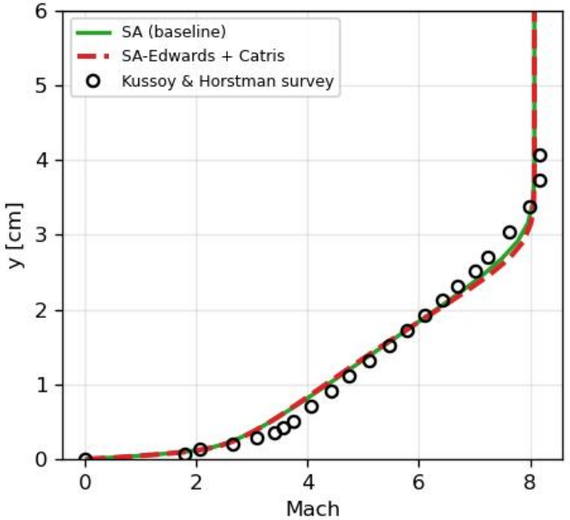



Mach 8.2 flat plate, incoming boundary layer at x = -11 cm: SA-Edwards + Catris vs SA vs experiment
SA-Edwards combined with the Catris-Aupoix density correction

linear height (whole layer)



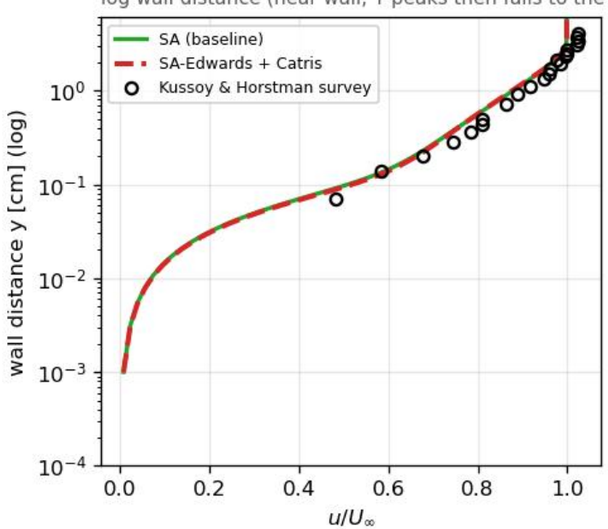



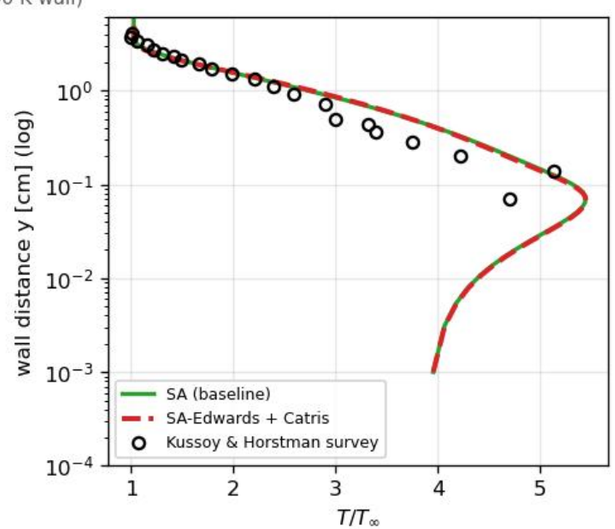



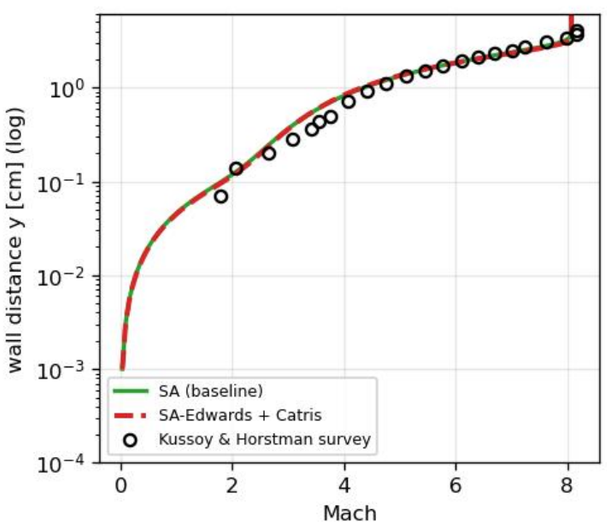

x = -11 cm; model = SA-Edwards-Catris
(theta matches for SA by construction)

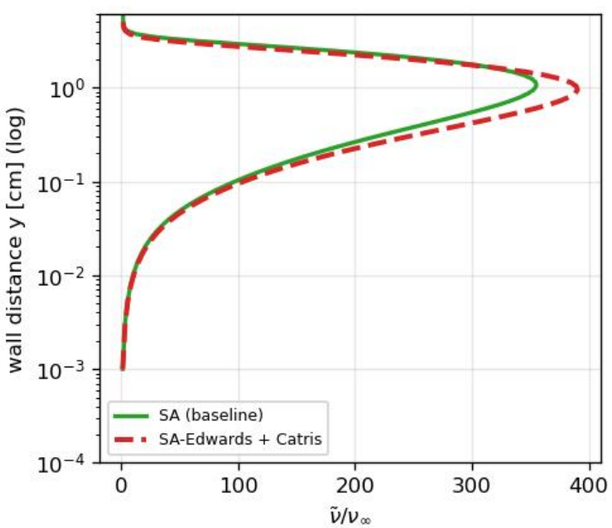
| quantity | exp. | SA | model | model vs SA | model vs exp. |
|---|---|---|---|---|---|
| theta, cm | 0.0940 | 0.0942 | 0.0928 | -1.4% | -1.2% |
| d99, cm | - | 2.635 | 2.555 | -3.0% | - |
| Cf x 1e3 | 0.980 | 0.953 | 0.969 | +1.7% | -1.1% |
| q_w, W/cm2 | 0.990 | 1.102 | 1.122 | +1.8% | +13.3% |

log wall distance (near wall; T peaks then falls to the 300 K wall)



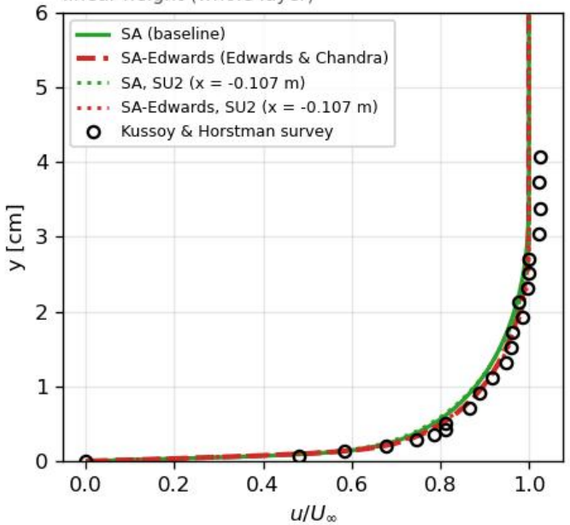



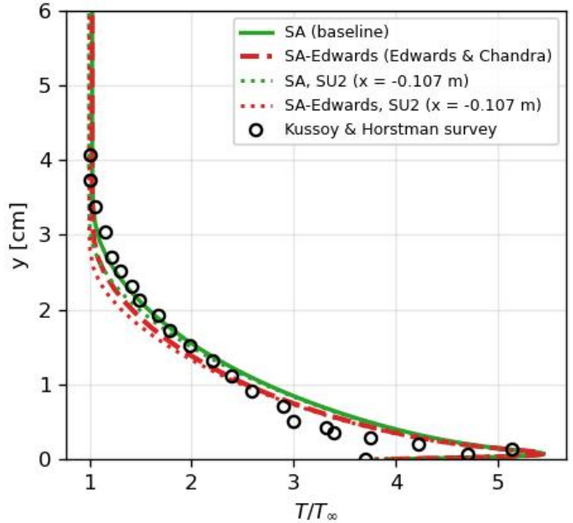



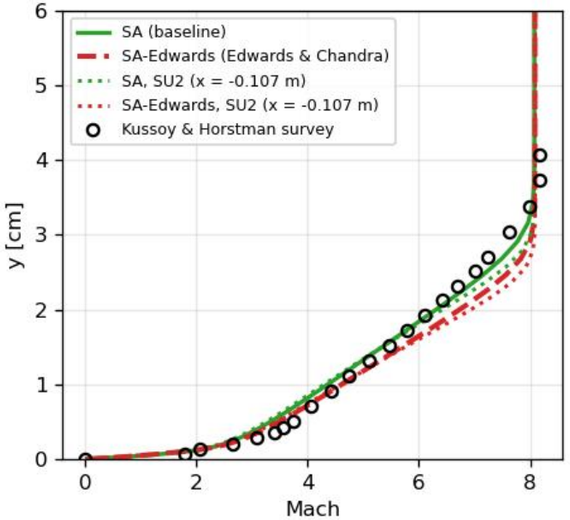



Mach 8.2 flat plate, incoming boundary layer at x = -11 cm:  SA-Edwards (Edwards & Chandra)  vs  SA  vs  experiment
Edwards & Chandra (AIAA 94-2275): strain-based production, Shat = S (1/chi + fv1), r = tanh(nuTilde/(Shat k^2 d^2))/tanh(1), no ft2

linear height (whole layer)



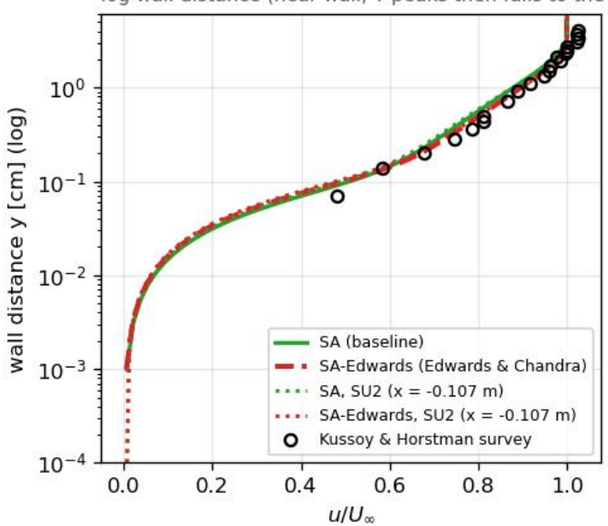



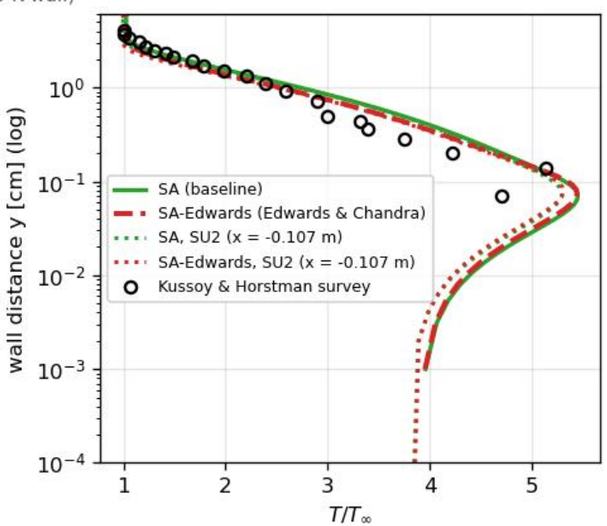



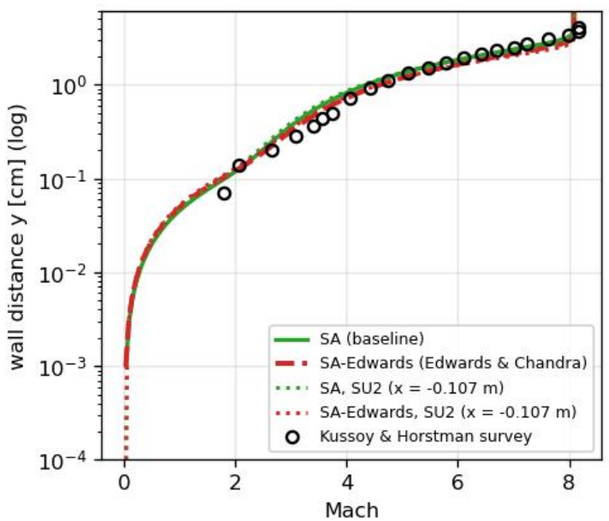

x = -11 cm; model = SA-Edwards
(theta matches for SA by construction)

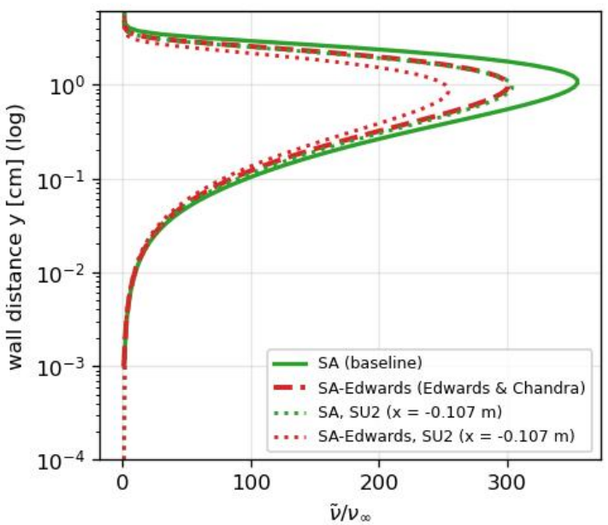
| quantity | exp. | SA | model | model vs SA | model vs exp. |
|---|---|---|---|---|---|
| theta, cm | 0.0940 | 0.0942 | 0.0835 | -11.4% | -11.2% |
| d99, cm | - | 2.635 | 2.376 | -9.8% | - |
| Cf x 1e3 | 0.980 | 0.953 | 0.881 | -7.5% | -10.1% |
| q_w, W/cm2 | 0.990 | 1.102 | 1.017 | -7.7% | +2.7% |

log wall distance (near wall; T peaks then falls to the 300 K wall)



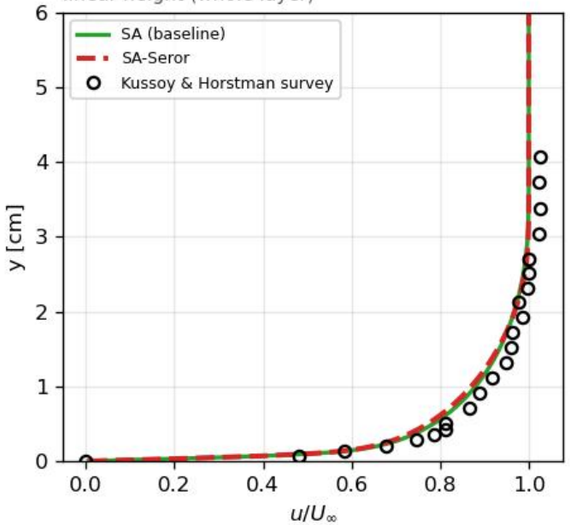



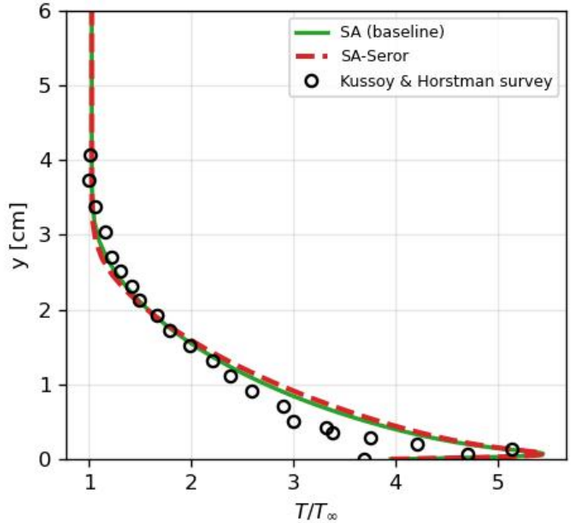



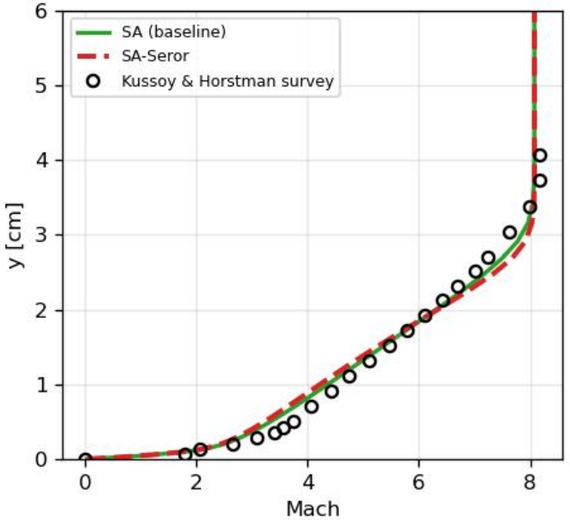



Mach 8.2 flat plate, incoming boundary layer at x = -11 cm: SA-Seror vs SA vs experiment
Seror et al. (2005) form of the density-gradient diffusion terms

linear height (whole layer)



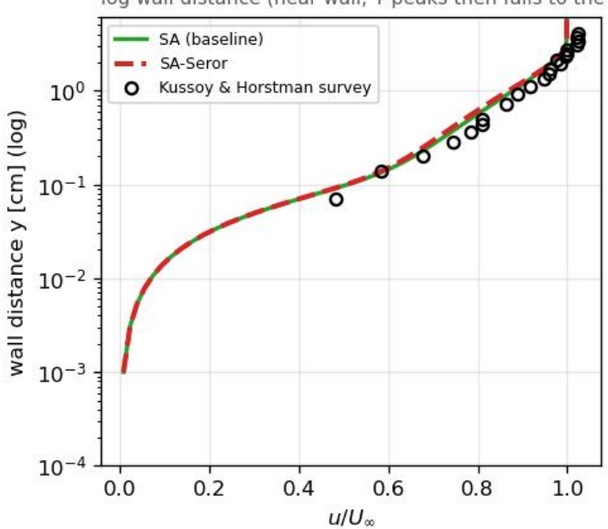



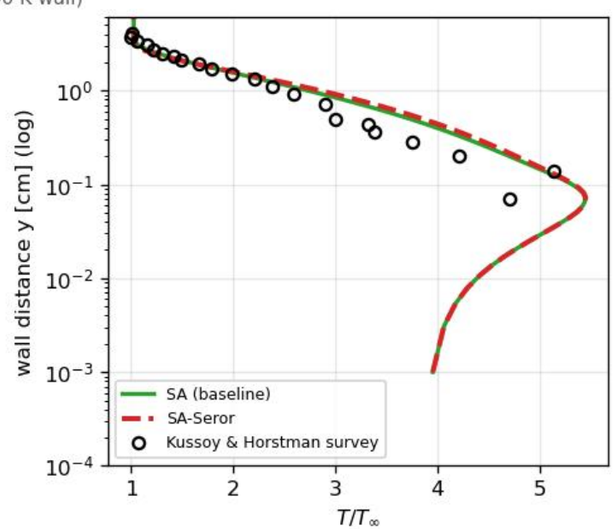



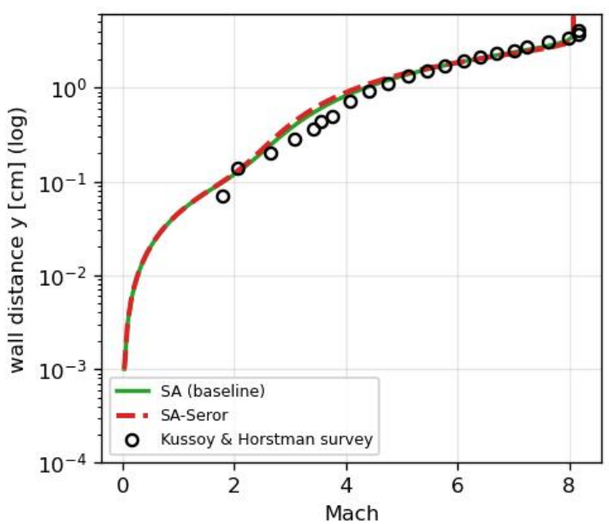

x = -11 cm; model = SA-Seror
(theta matches for SA by construction)

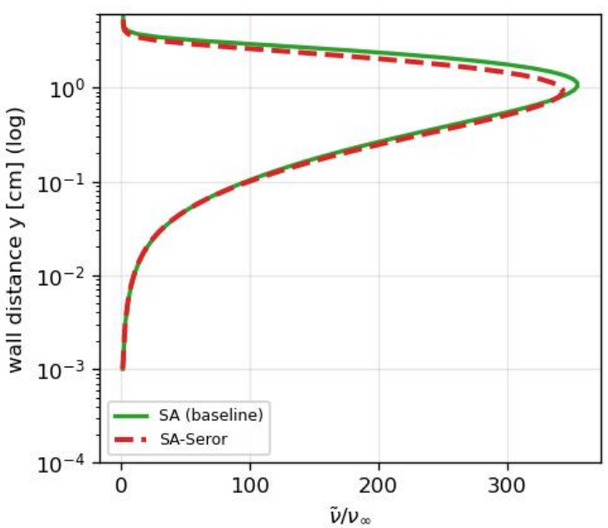
| quantity | exp. | SA | model | model vs SA | model vs exp. |
|---|---|---|---|---|---|
| theta, cm | 0.0940 | 0.0942 | 0.0932 | -1.0% | -0.8% |
| d99, cm | - | 2.635 | 2.527 | -4.1% | - |
| Cf x 1e3 | 0.980 | 0.953 | 0.950 | -0.3% | -3.0% |
| q_w, W/cm2 | 0.990 | 1.102 | 1.101 | -0.1% | +11.2% |

log wall distance (near wall; T peaks then falls to the 300 K wall)



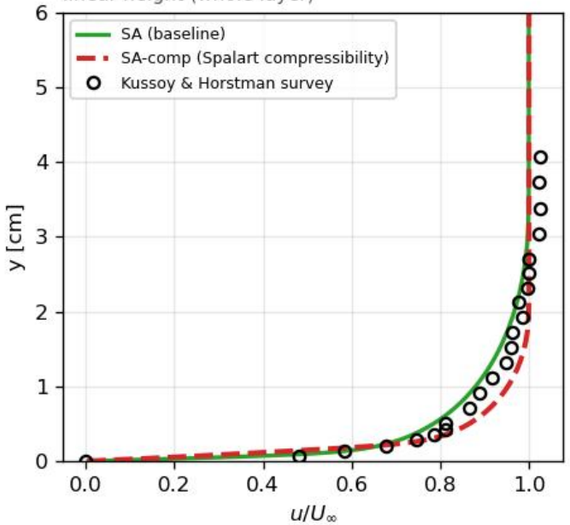



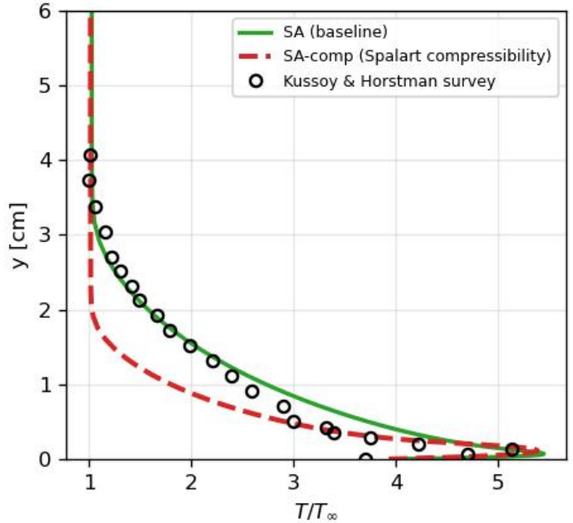



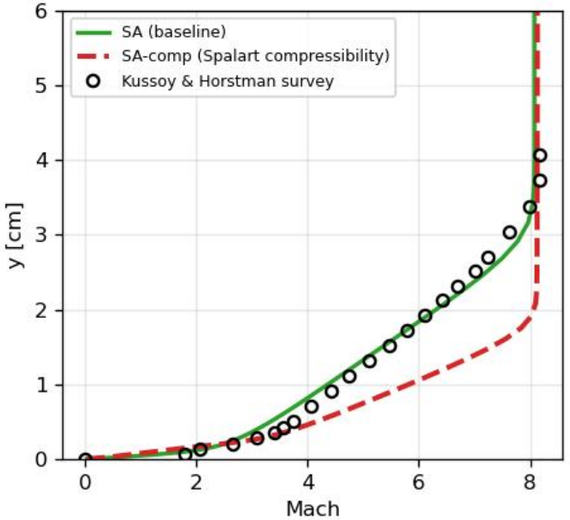



Mach 8.2 flat plate, incoming boundary layer at x = -11 cm:  SA-comp (Spalart compressibility)  vs  SA  vs  experiment
Spalart compressibility correction (AIAA 2000-2306): extra destruction -C5 nuTilde^2 (du_i/dx_j)^2 / a^2, C5 = 3.5

linear height (whole layer)



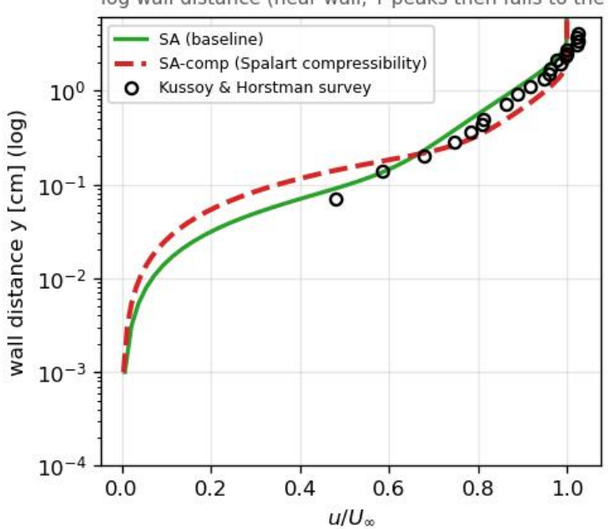



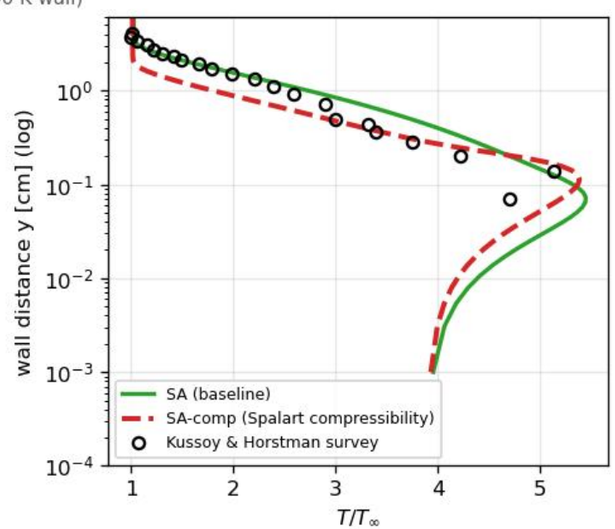



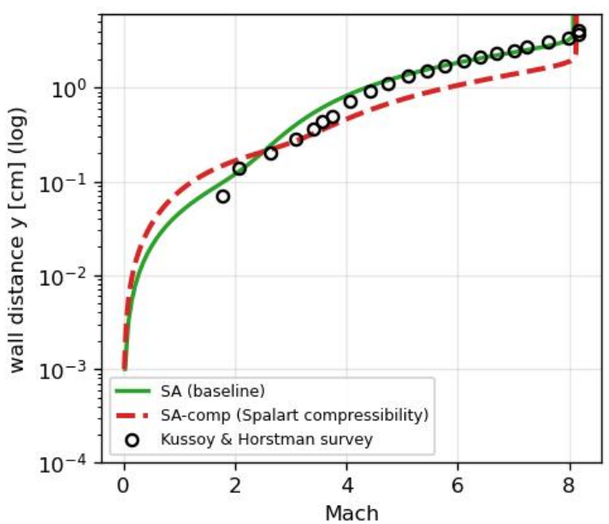

x = -11 cm; model = SA-comp
(theta matches for SA by construction)

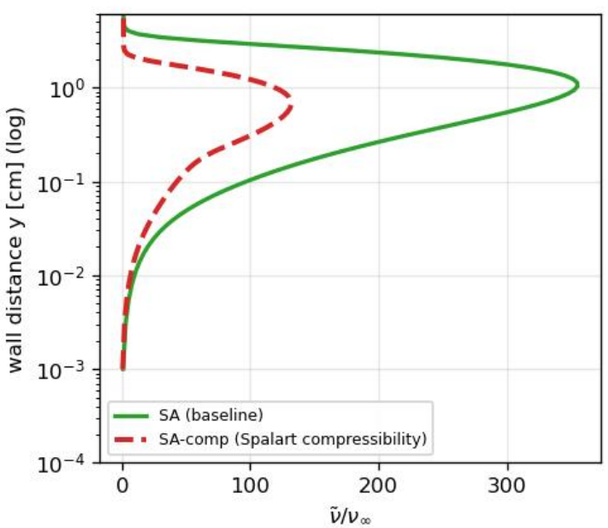
| quantity | exp. | SA | model | model vs SA | model vs exp. |
|---|---|---|---|---|---|
| theta, cm | 0.0940 | 0.0942 | 0.0508 | -46.0% | -45.9% |
| d99, cm | - | 2.635 | 1.548 | -41.3% | - |
| Cf x 1e3 | 0.980 | 0.953 | 0.551 | -42.2% | -43.8% |
| q_w, W/cm2 | 0.990 | 1.102 | 0.625 | -43.3% | -36.9% |

log wall distance (near wall; T peaks then falls to the 300 K wall)



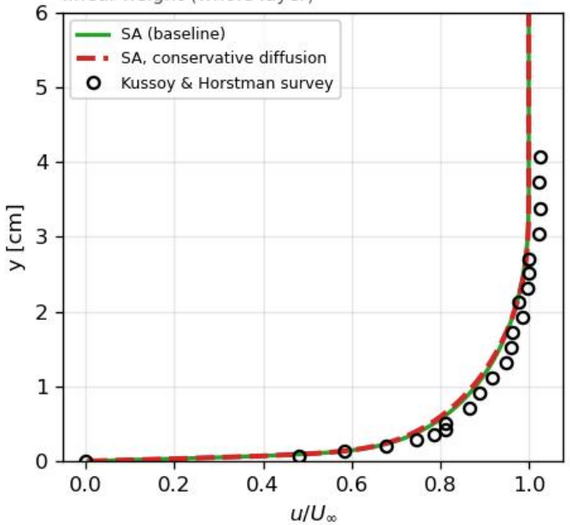



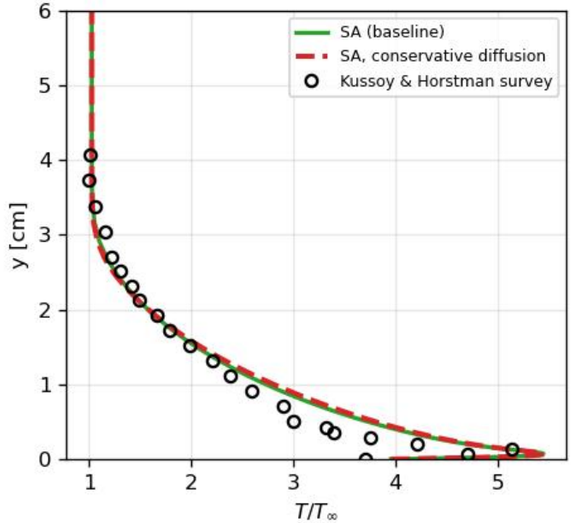



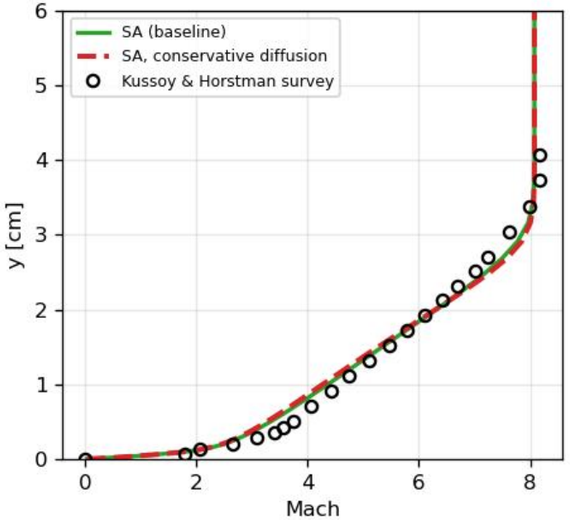



Mach 8.2 flat plate, incoming boundary layer at x = -11 cm:  SA, conservative diffusion  vs  SA  vs  experiment
conservative diffusion form (TMR / SU2): adds the density-gradient cross terms

linear height (whole layer)



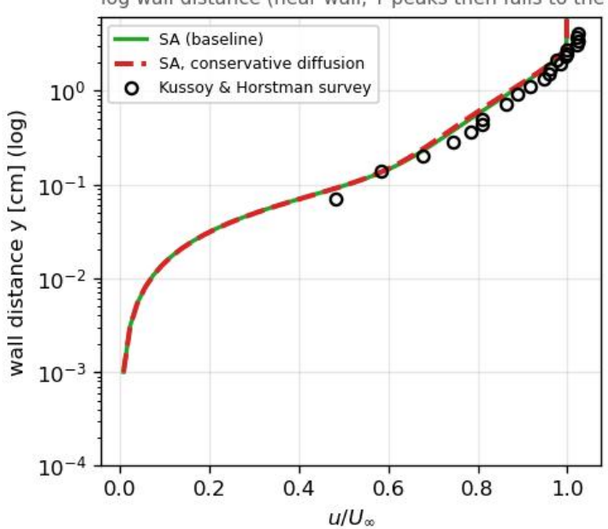



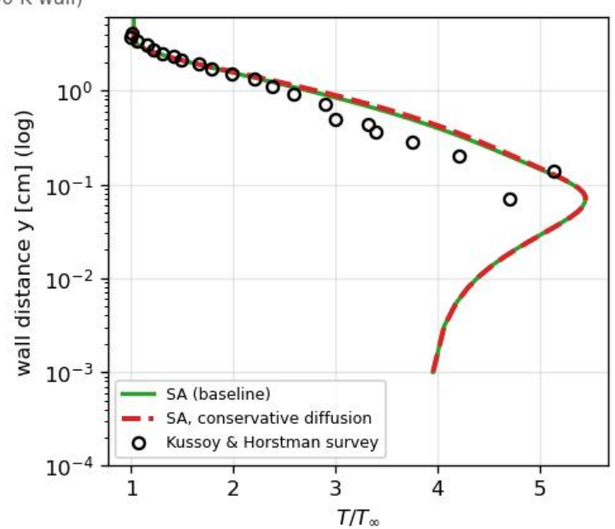



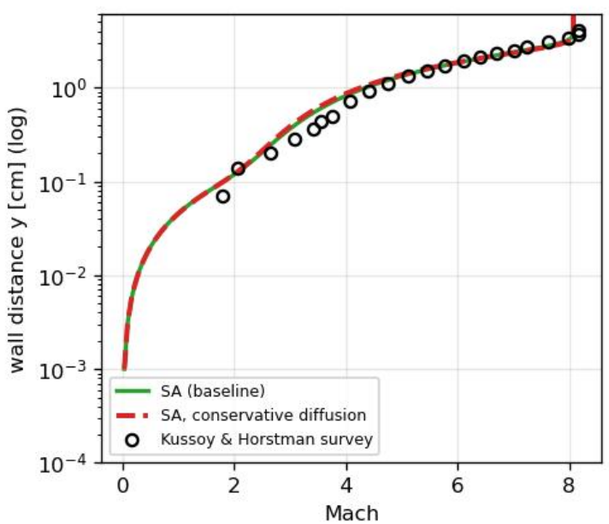

x = -11 cm; model = SA-cons
(theta matches for SA by construction)

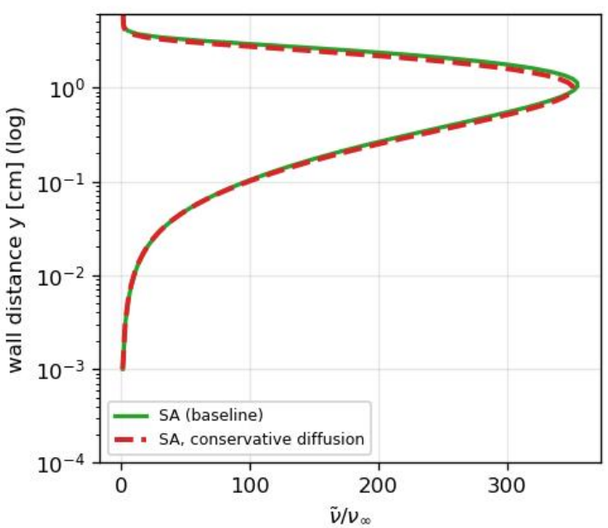
| quantity | exp. | SA | model | model vs SA | model vs exp. |
|---|---|---|---|---|---|
| theta, cm | 0.0940 | 0.0942 | 0.0941 | -0.1% | +0.1% |
| d99, cm | - | 2.635 | 2.586 | -1.9% | - |
| Cf x 1e3 | 0.980 | 0.953 | 0.956 | +0.4% | -2.4% |
| q_w, W/cm2 | 0.990 | 1.102 | 1.108 | +0.5% | +11.9% |

log wall distance (near wall; T peaks then falls to the 300 K wall)

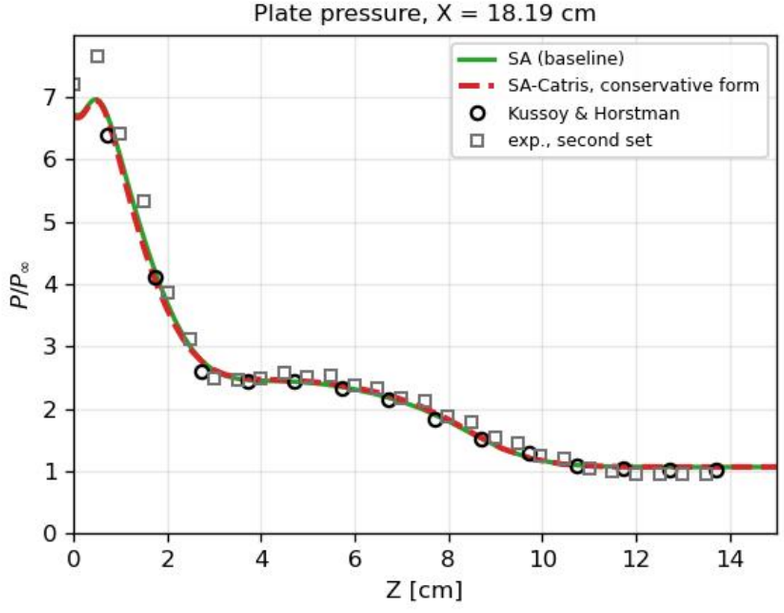


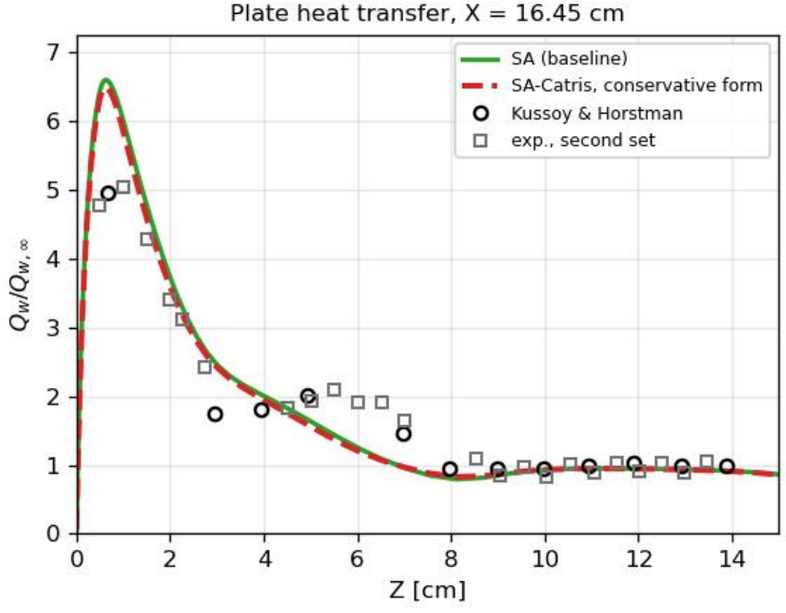



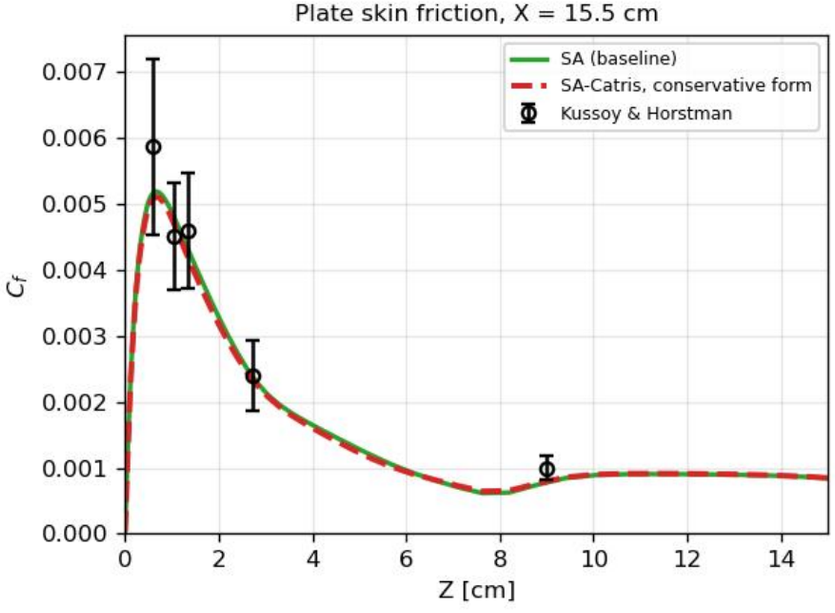



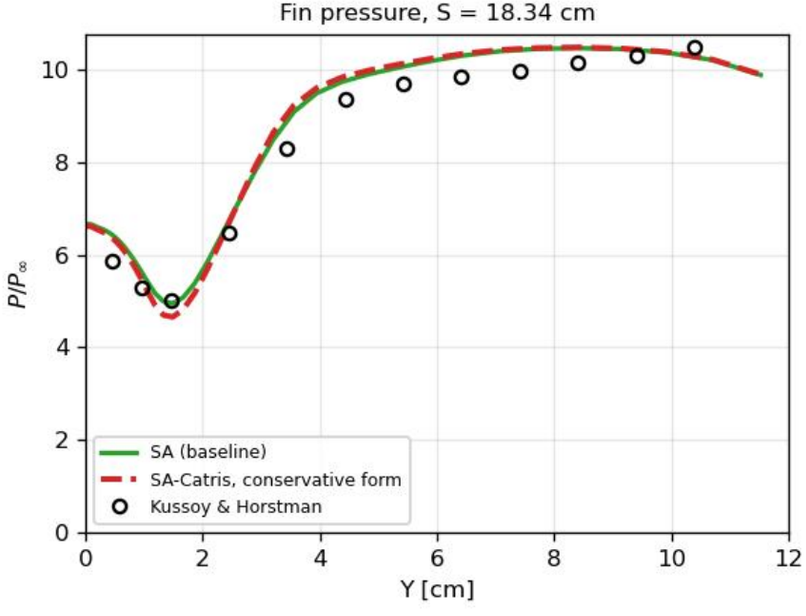



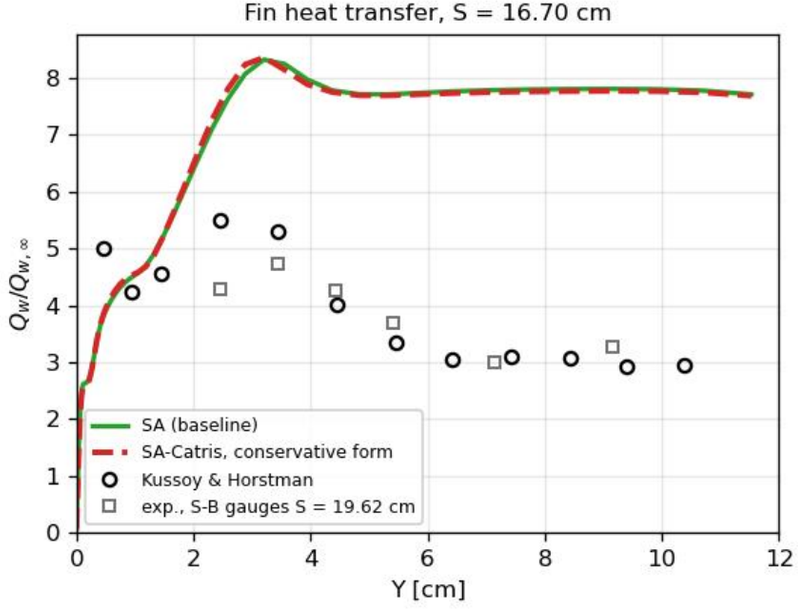
Kussoy & Horstman Mach 8.2, 15 deg sharp fin: SA-Catris, conservative form vs SA vs experiment
Catris-Aupoix density correction, conservative left-hand side (TMR SA-noft2-CatrisCons)



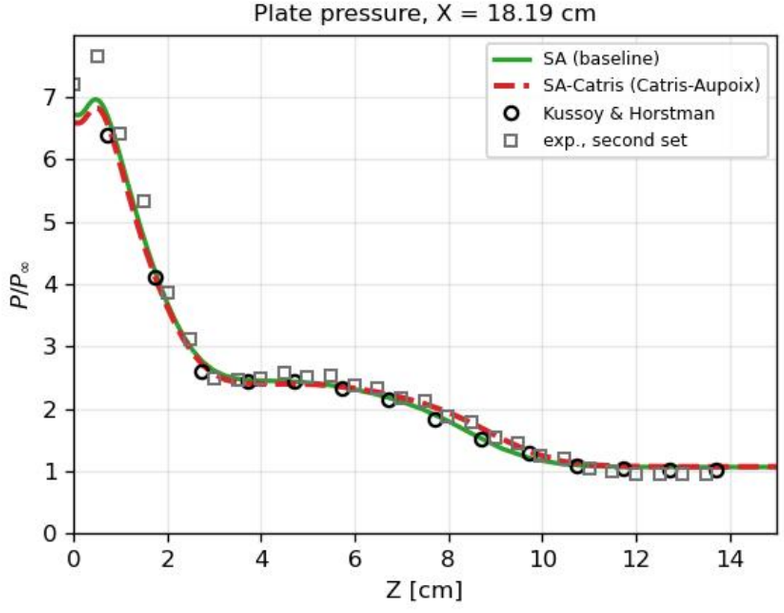



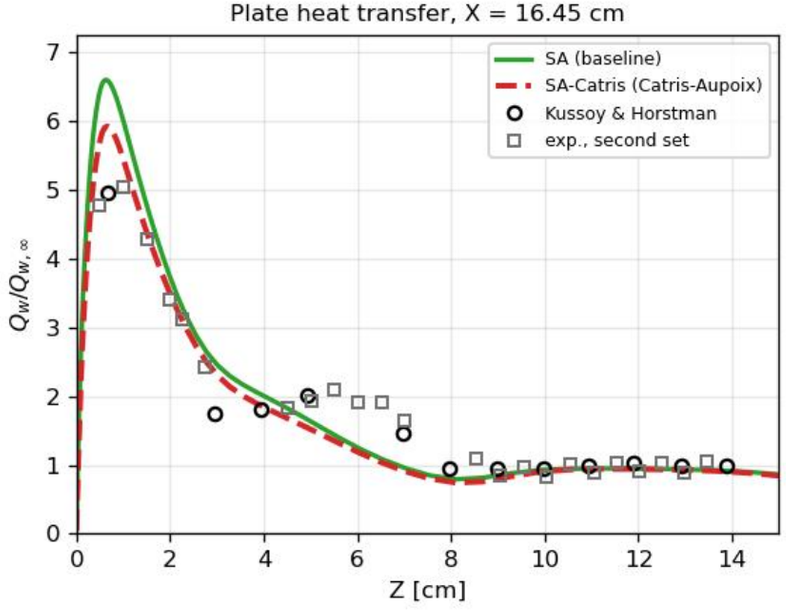



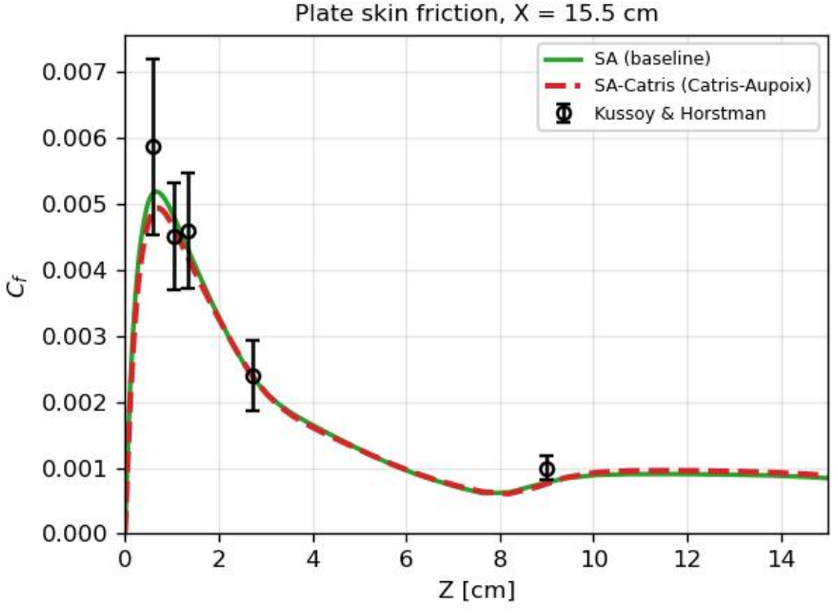



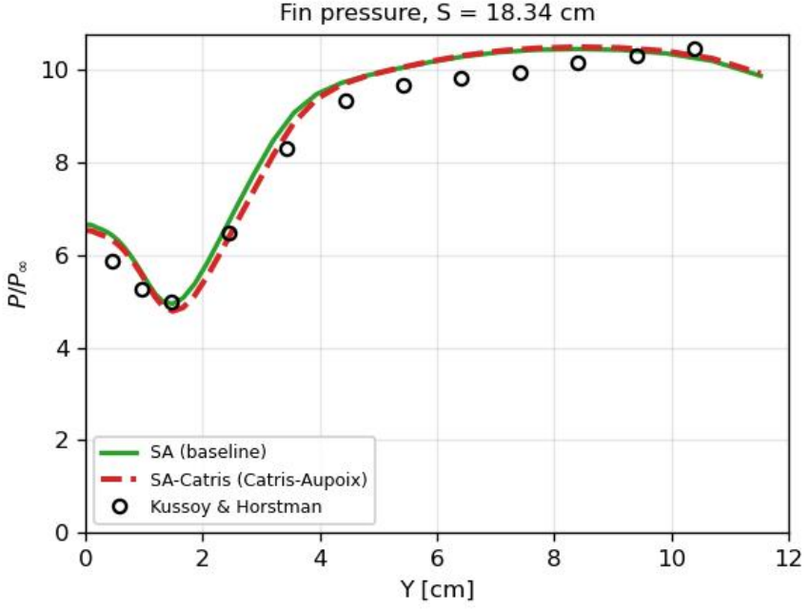



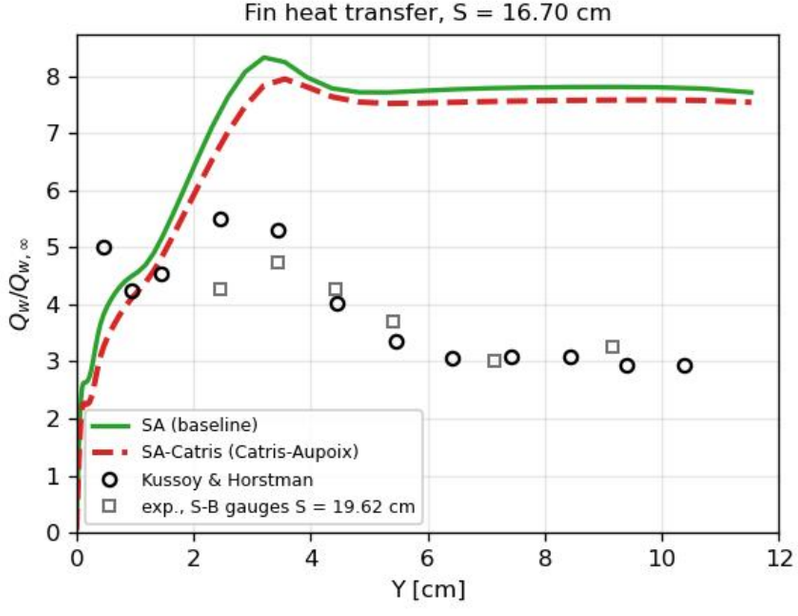
Kussoy & Horstman Mach 8.2, 15 deg sharp fin:  SA-Catris (Catris-Aupoix)  vs  SA  vs  experiment
Catris-Aupoix density correction (2000): diffuses sqrt(rho) nuTilde, laminar diffusion of mu, original non-conservative left-hand side



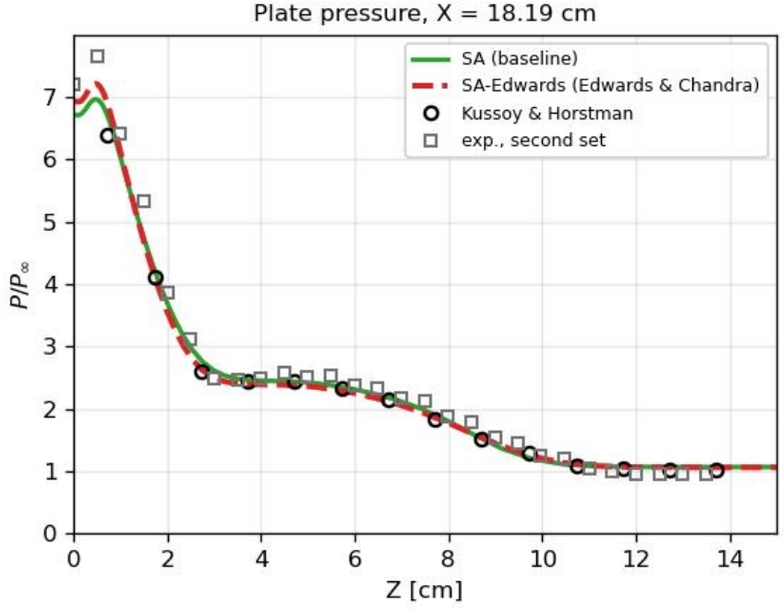



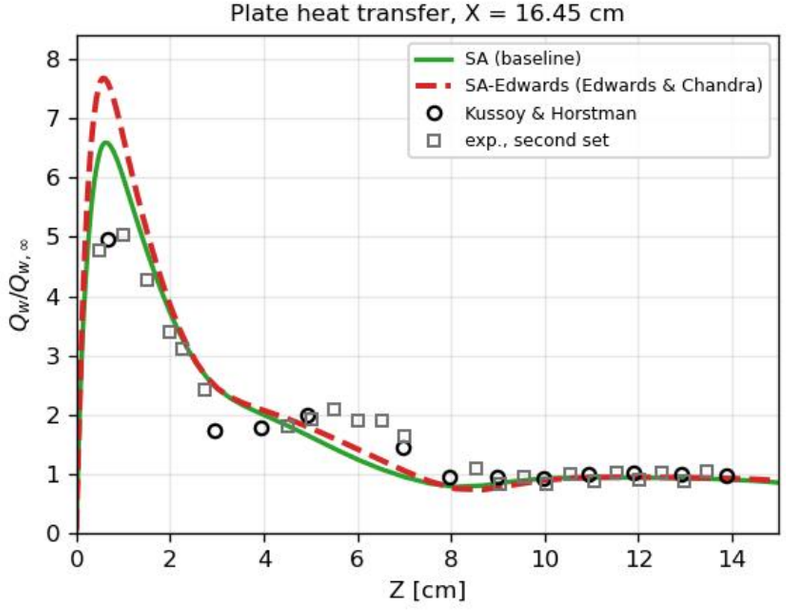



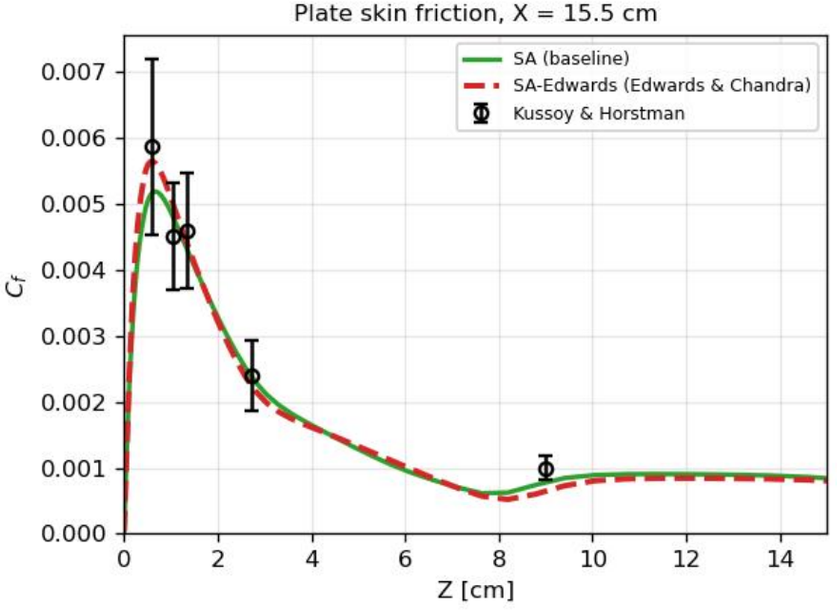



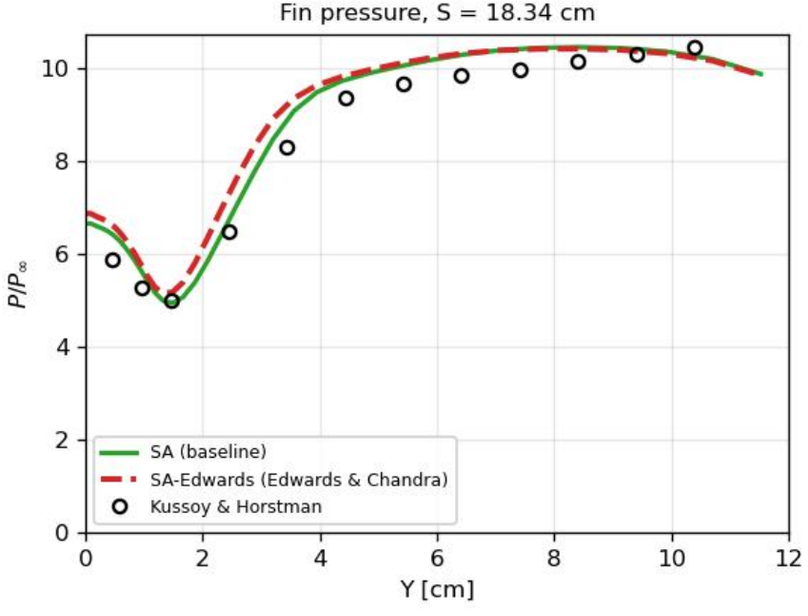



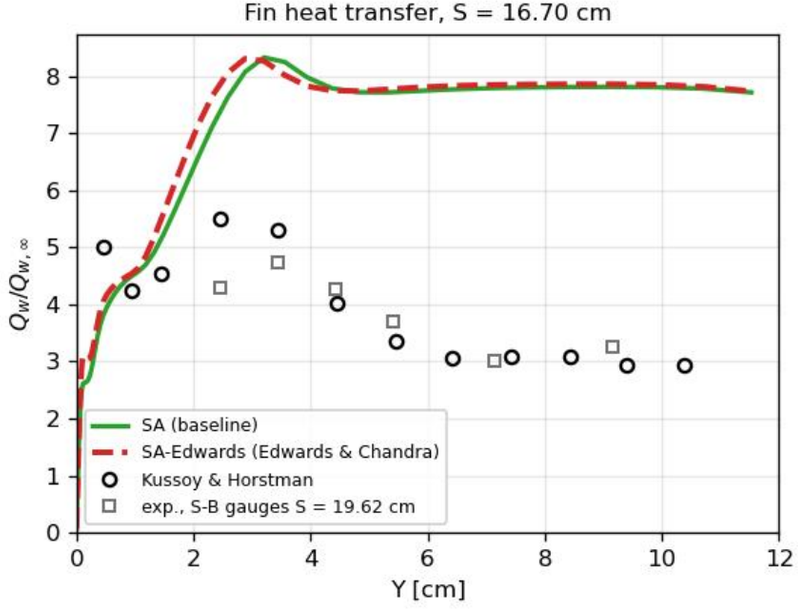
Kussoy & Horstman Mach 8.2, 15 deg sharp fin:  SA-Edwards (Edwards & Chandra)  vs  SA  vs  experiment
Edwards & Chandra (AIAA 94-2275): strain-based production, Shat = S (1/chi + fv1), r = tanh(nuTilde/(Shat k^2 d^2))/tanh(1), no ft2



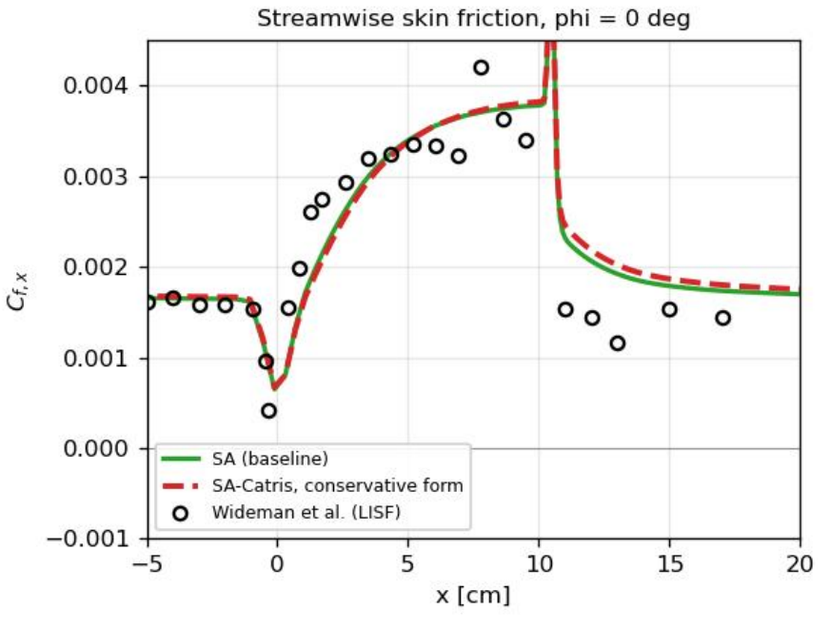



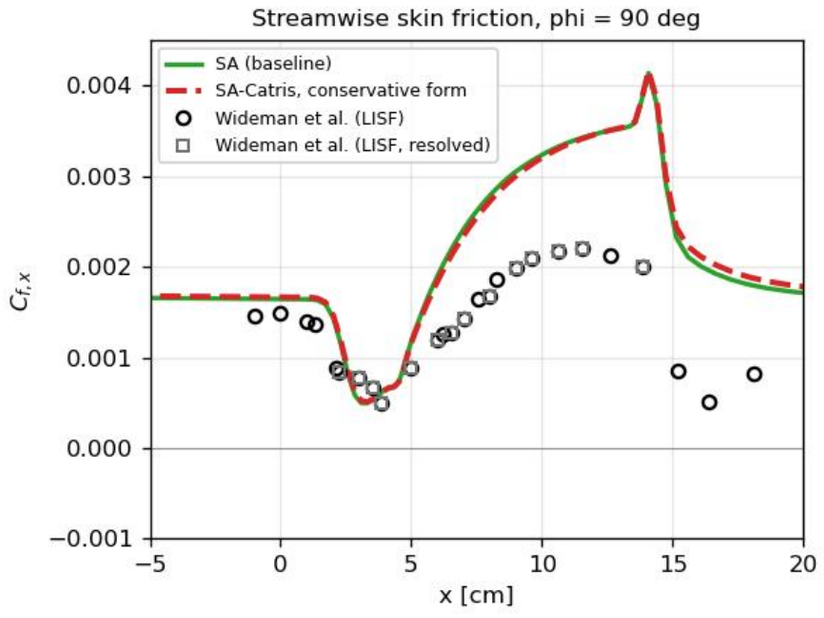



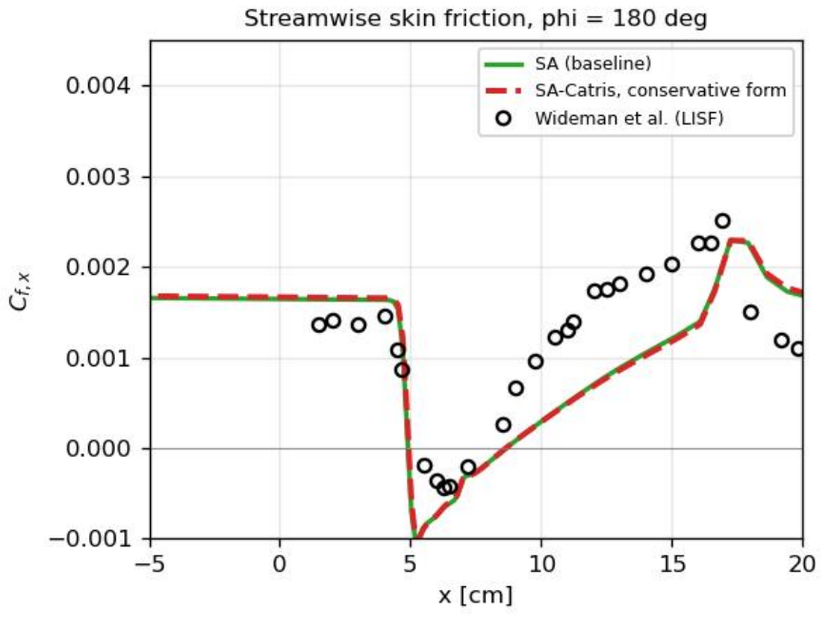



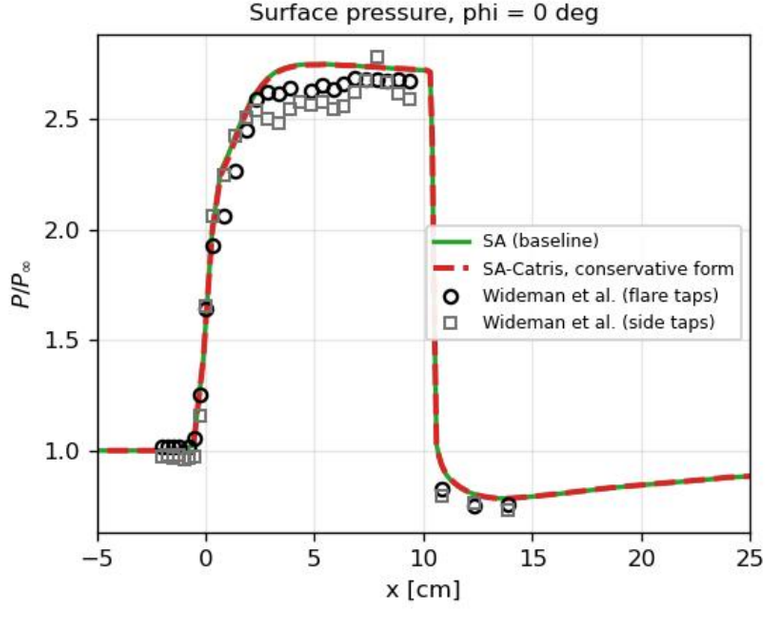



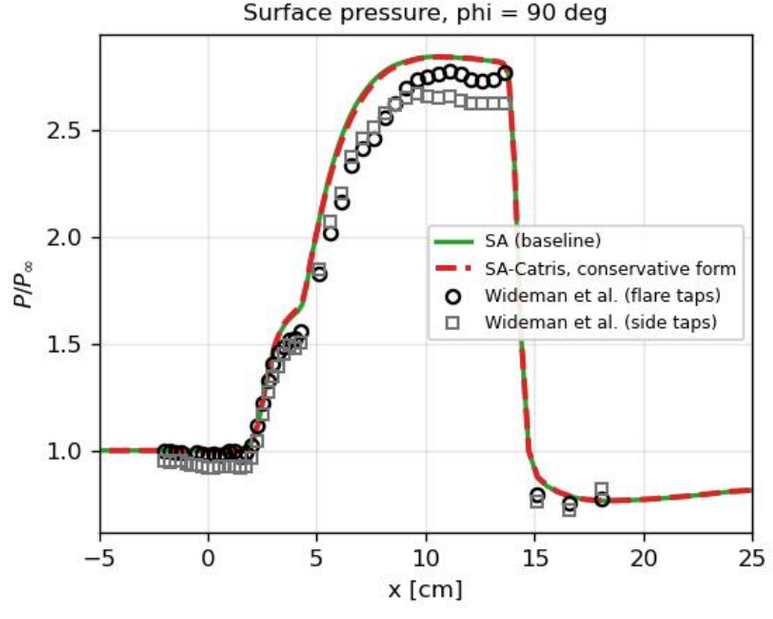
Wideman et al. Mach 2.89 cylinder / offset 20 deg flare: SA-Catris, conservative form vs SA vs experiment
Catris-Aupoix density correction, conservative left-hand side (TMR SA-noft2-CatrisCons)

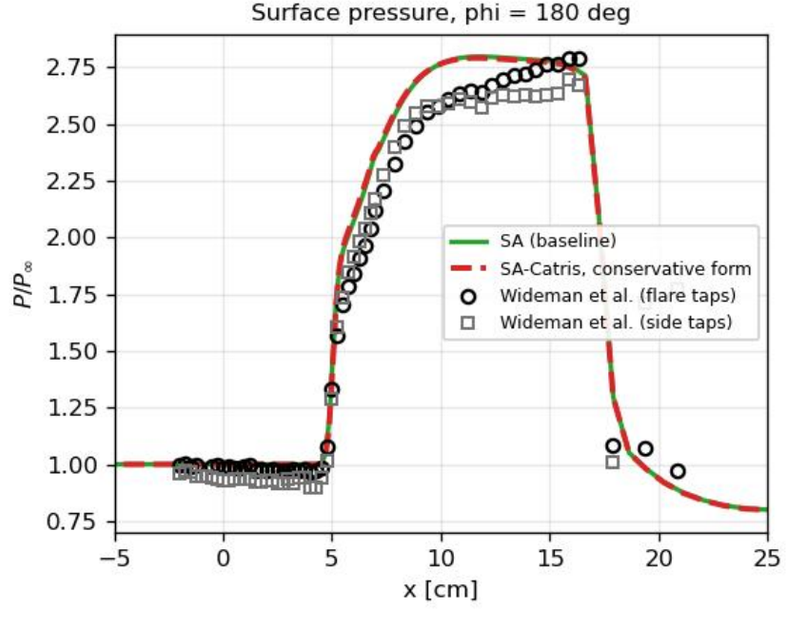


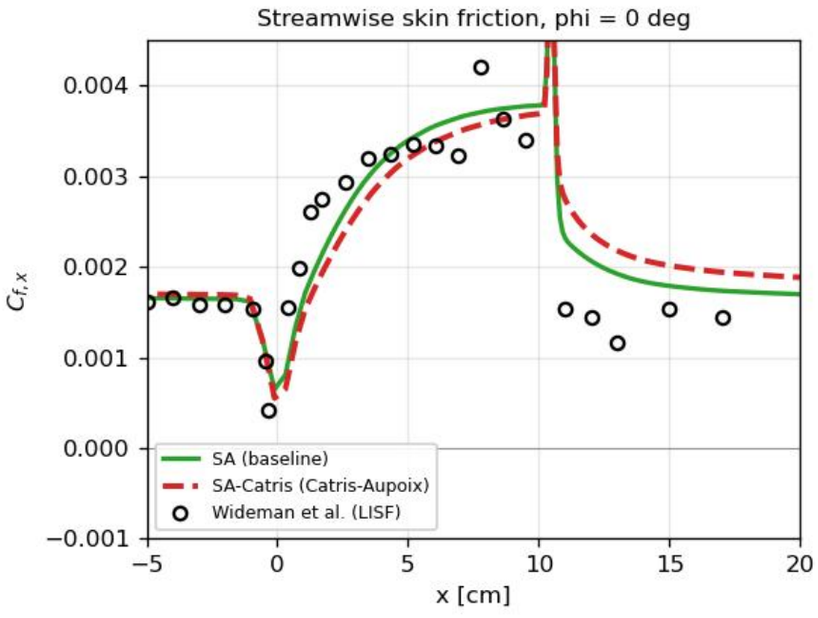



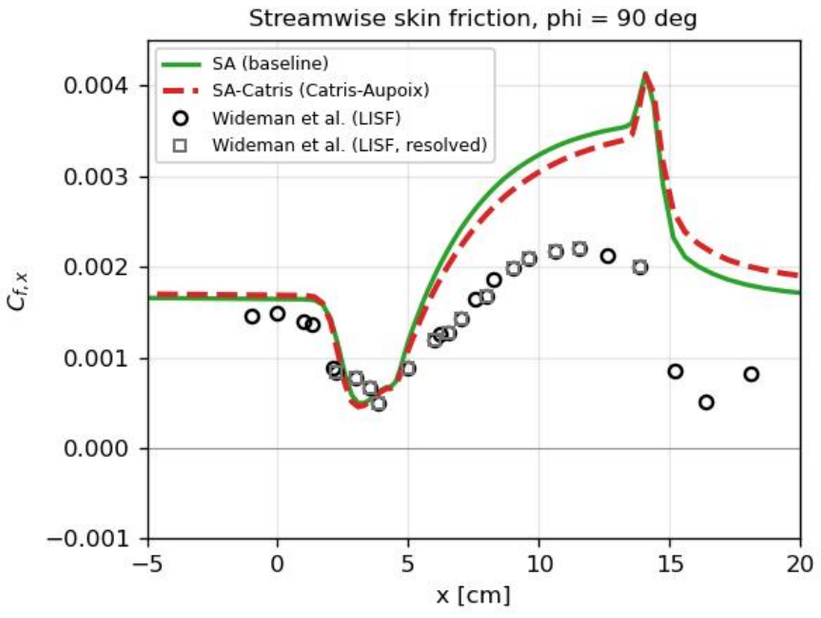



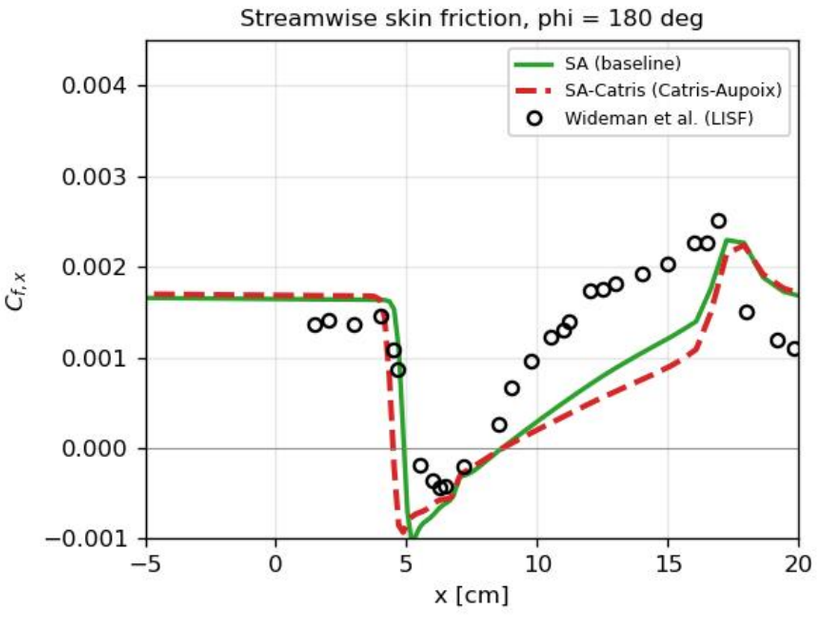



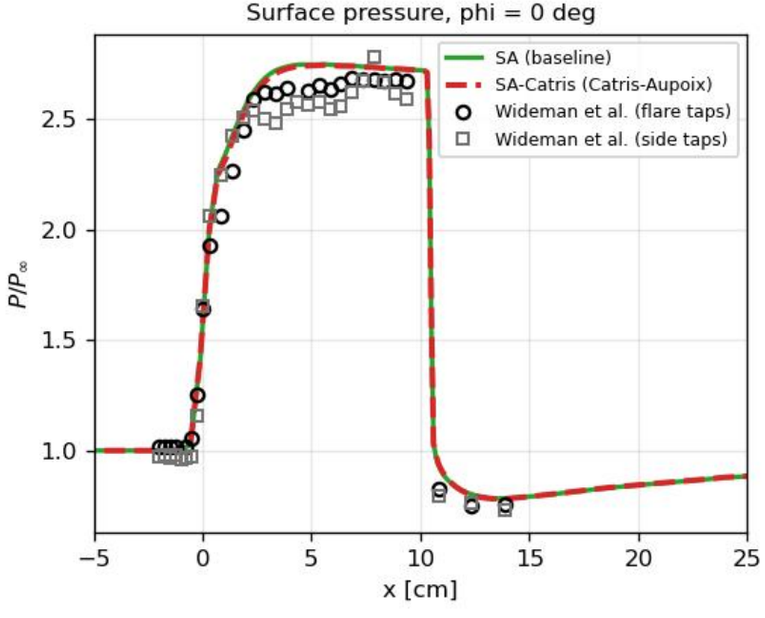



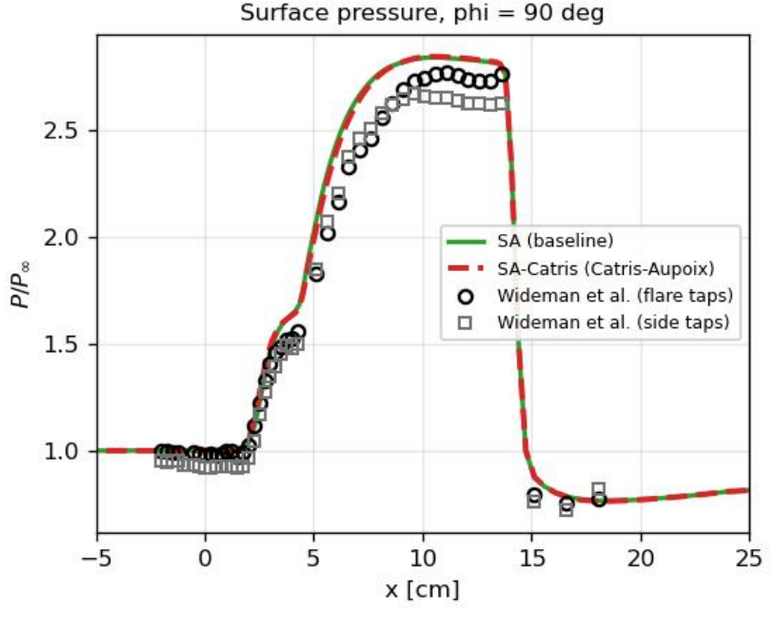
Wideman et al. Mach 2.89 cylinder / offset 20 deg flare: SA-Catris (Catris-Aupoix) vs SA vs experiment
Catris-Aupoix density correction (2000): diffuses sqrt(rho) nuTilde, laminar diffusion of mu, original non-conservative left-hand side

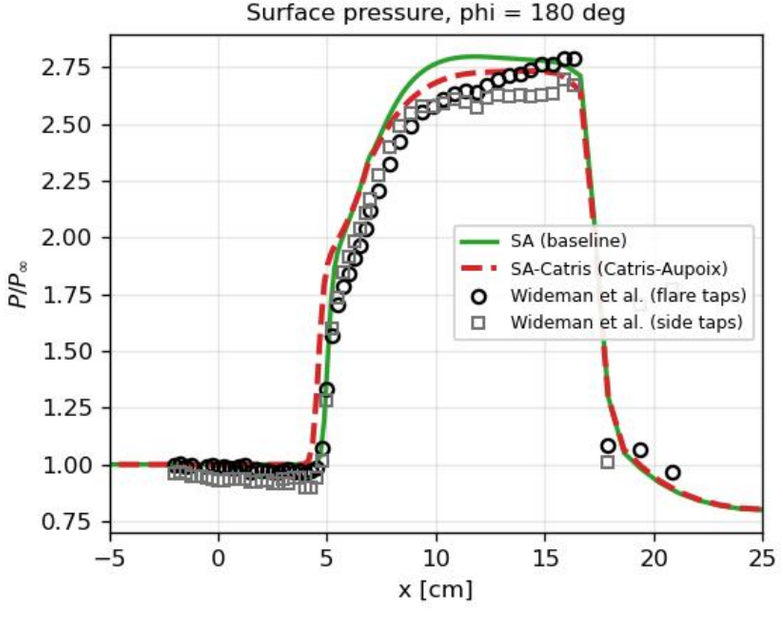


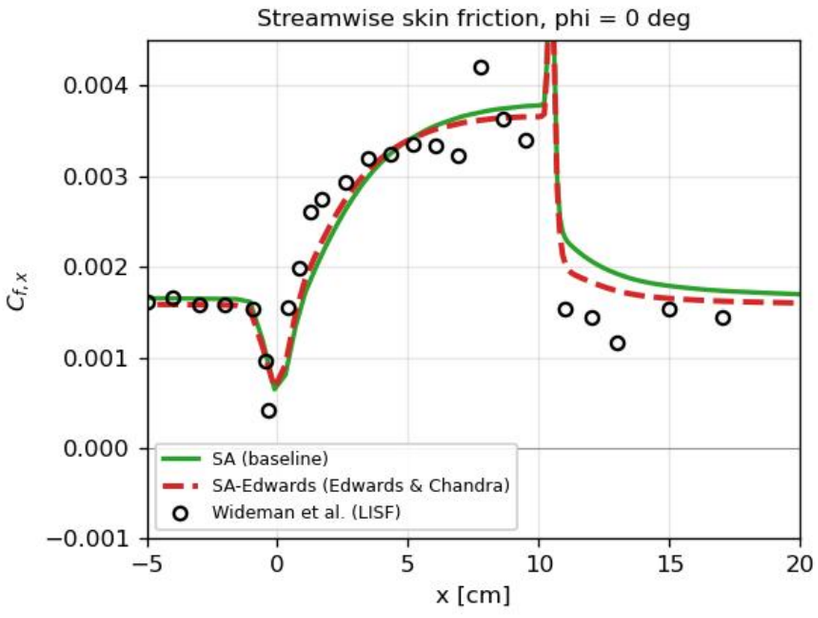



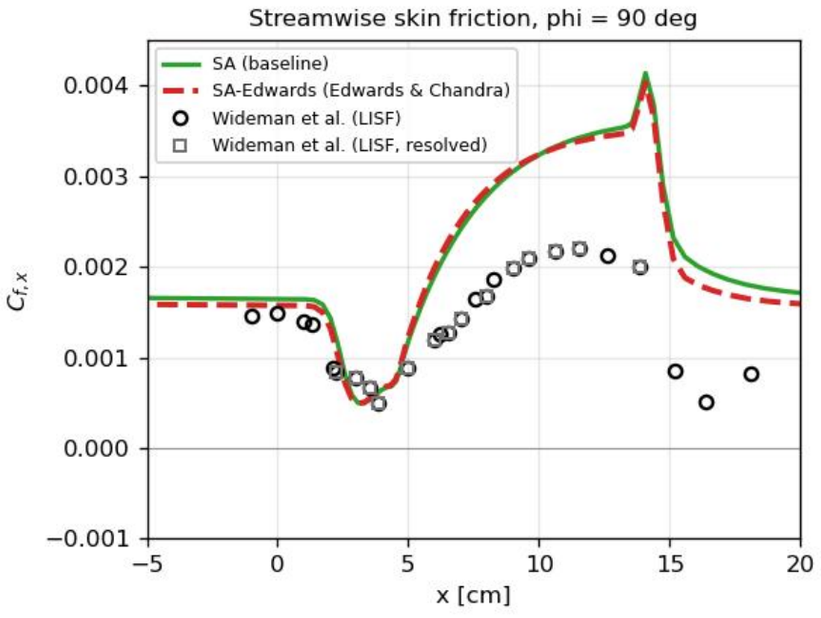



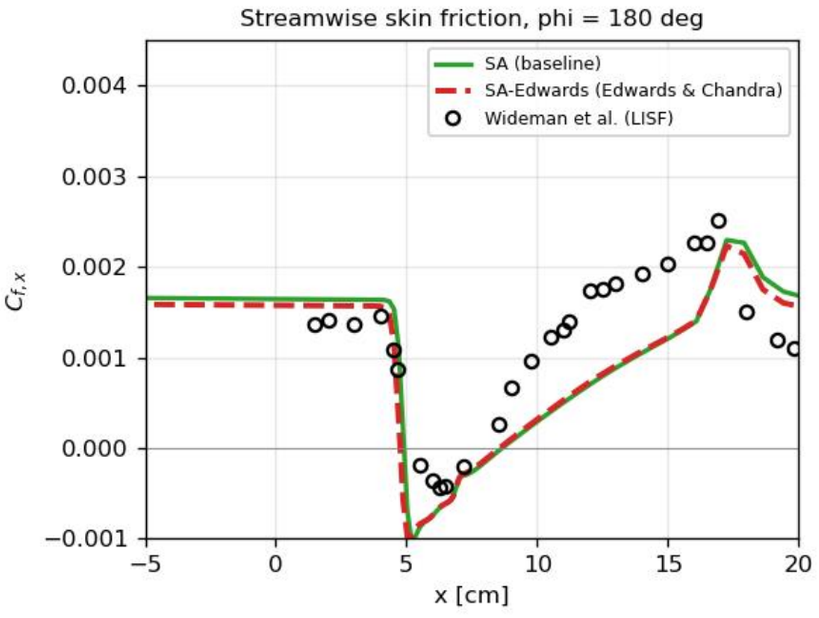



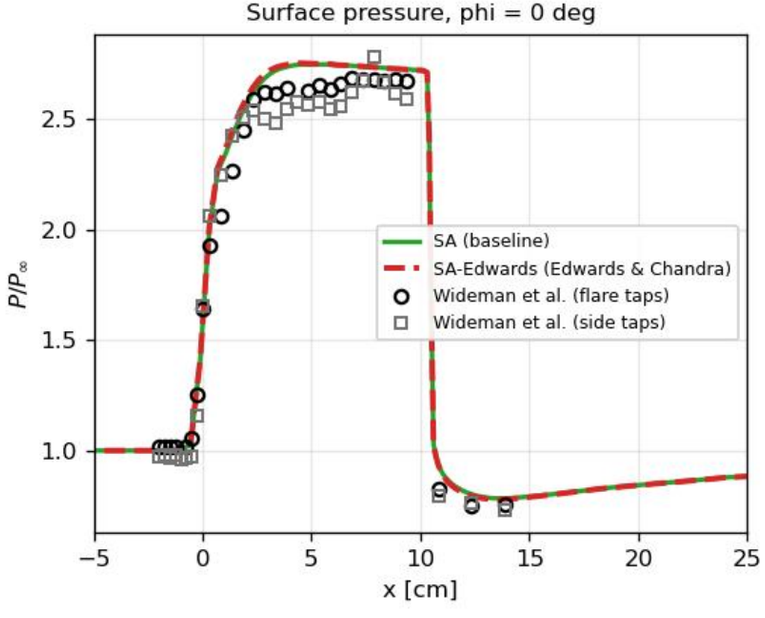



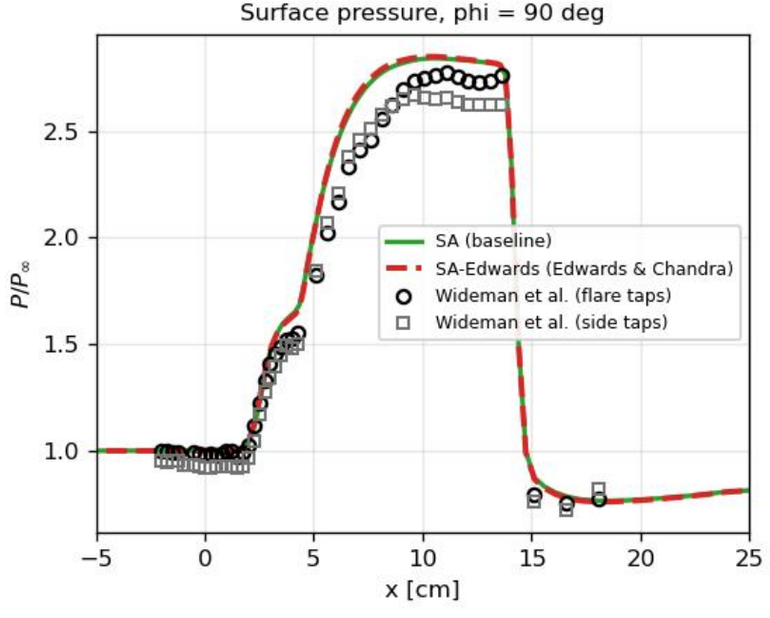
Wideman et al. Mach 2.89 cylinder / offset 20 deg flare:  SA-Edwards (Edwards & Chandra)  vs  SA  vs  experiment
Edwards & Chandra (AIAA 94-2275): strain-based production, Shat = S (1/chi + fv1), r = tanh(nuTilde/(Shat k^2 d^2))/tanh(1), no ft2

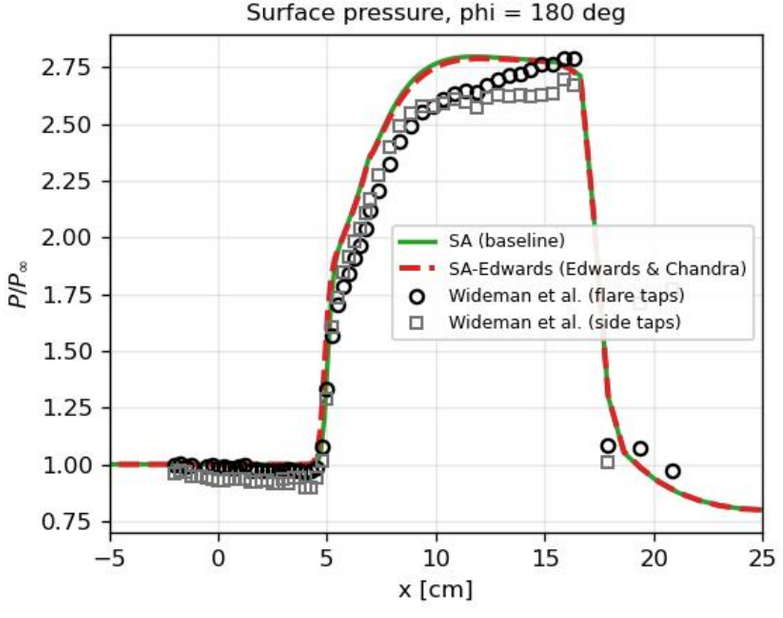


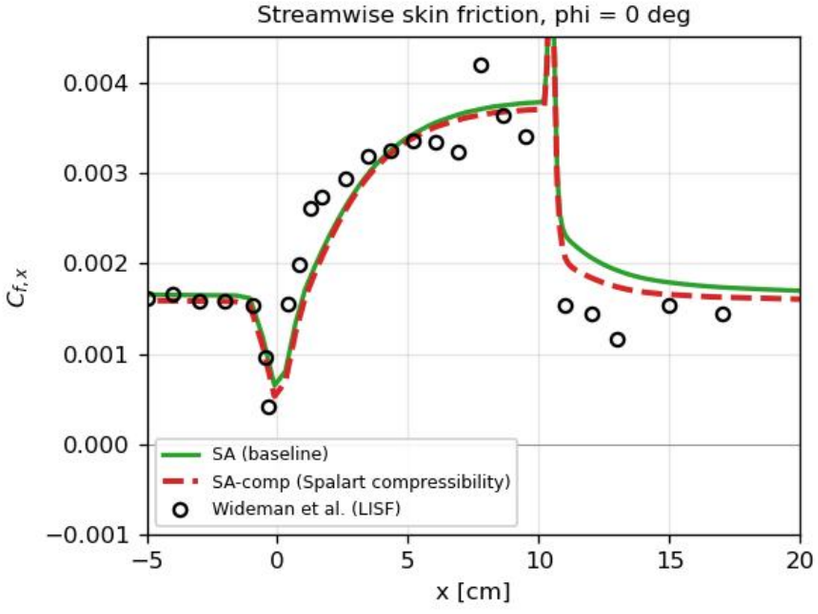



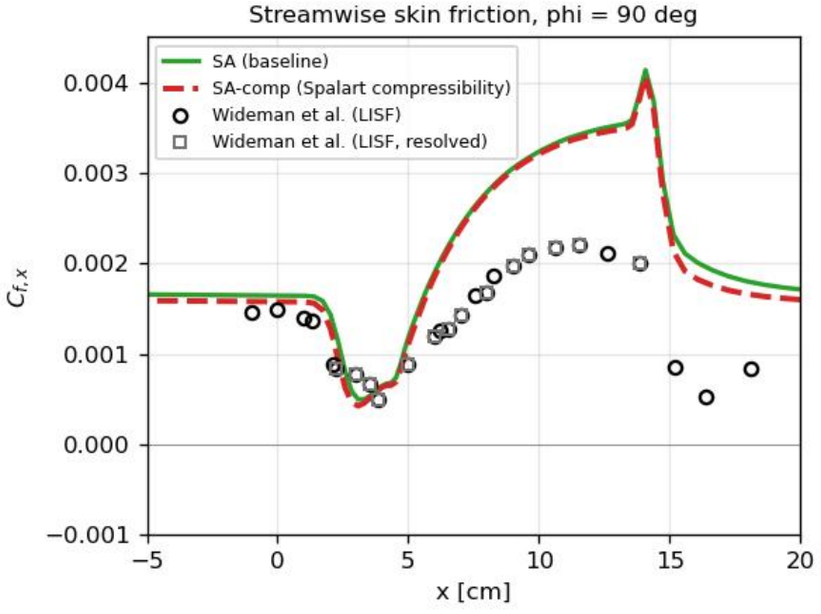



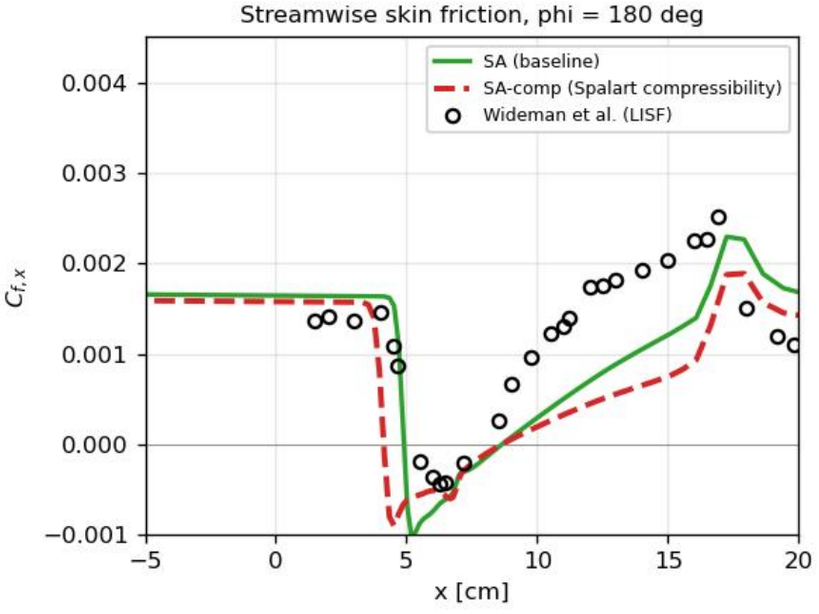



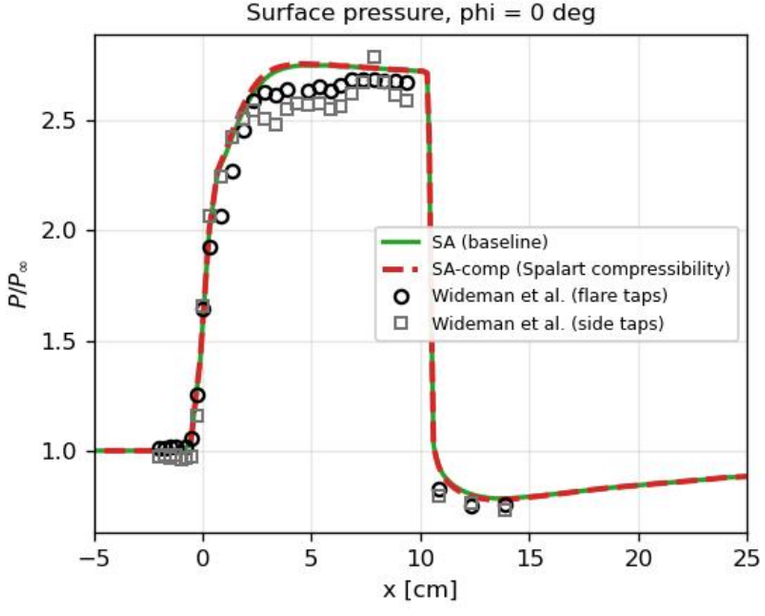



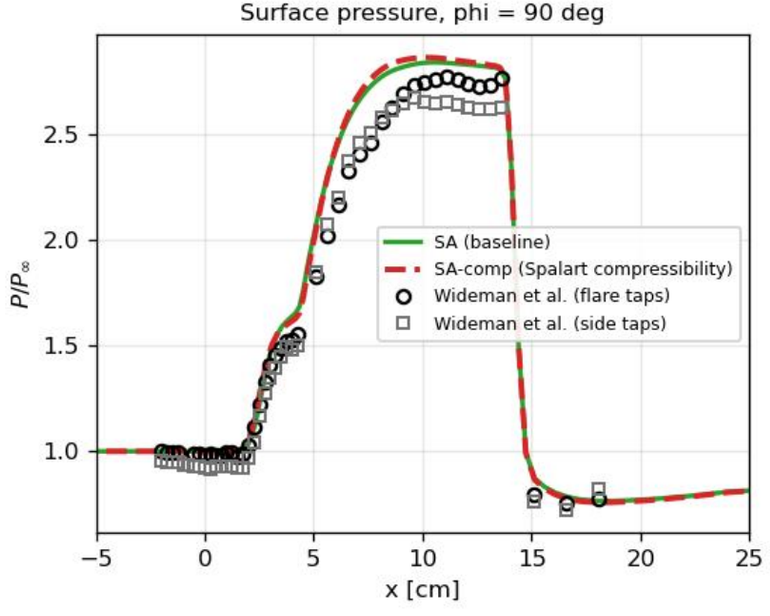
Wideman et al. Mach 2.89 cylinder / offset 20 deg flare: SA-comp (Spalart compressibility) vs SA vs experiment
Spalart compressibility correction (AIAA 2000-2306): extra destruction -C5 nuTilde^2 (du_i/dx_j)^2 / a^2, C5 = 3.5

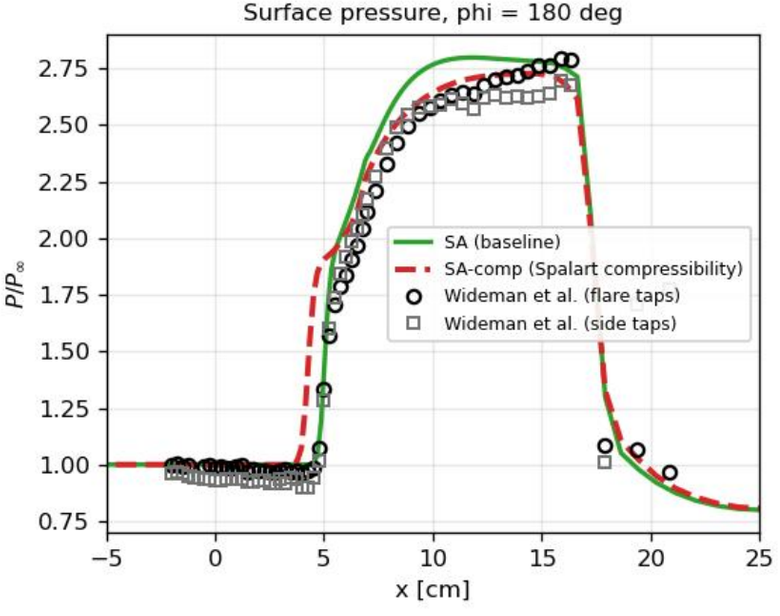


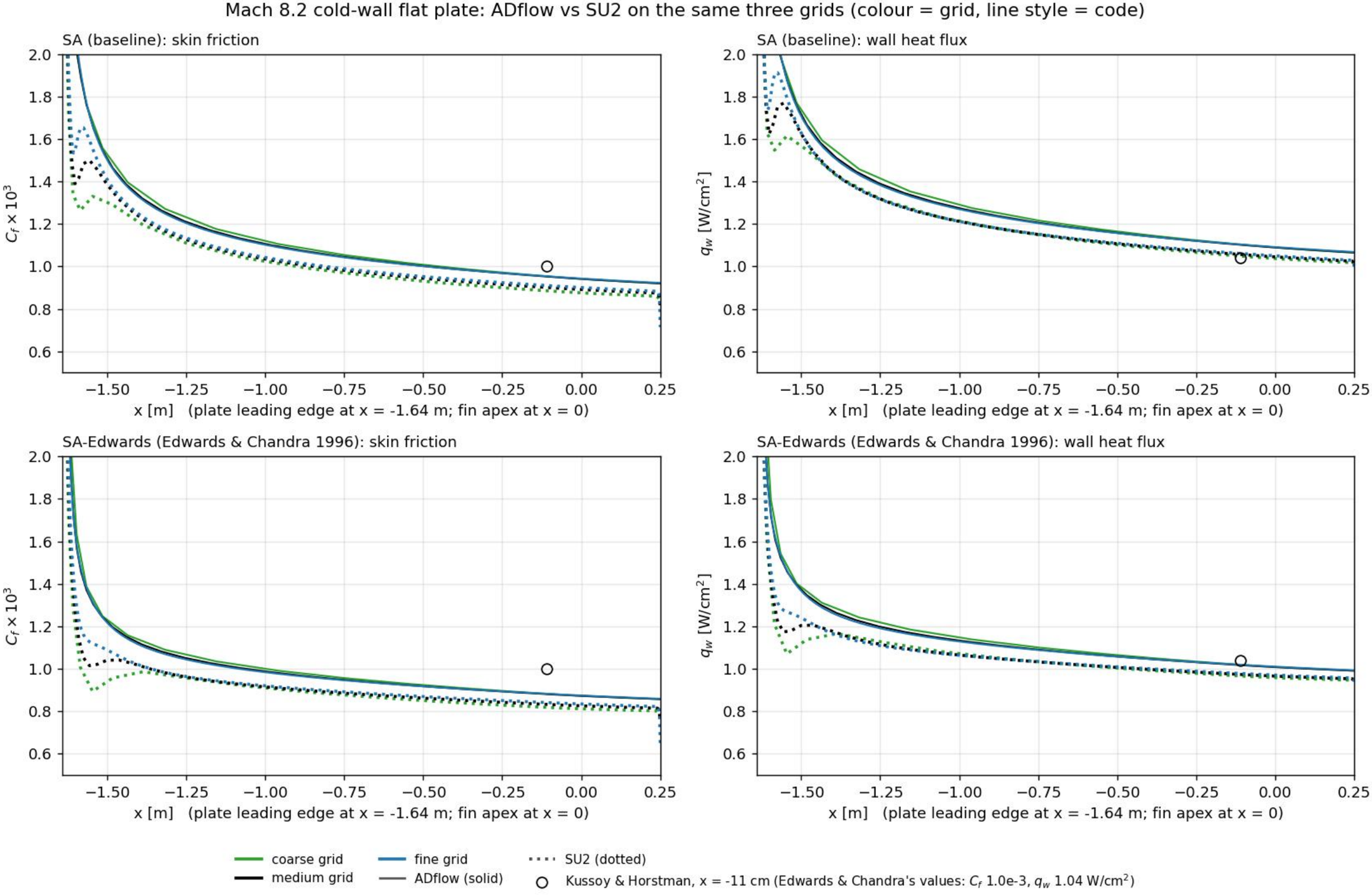



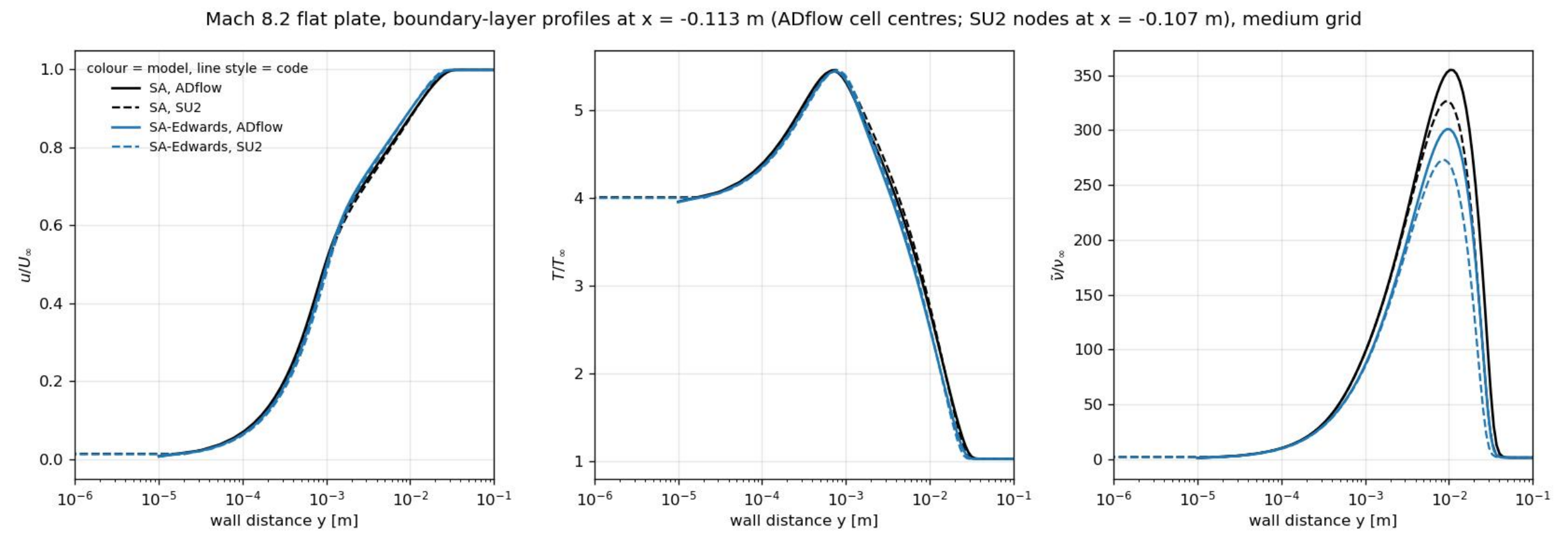




Mach 8.2 cold-wall flat plate: ADflow vs SU2 on the same three grids (colour = grid, line style = code)
SA (baseline): skin friction
SA (baseline): wall heat flux
SA-Edwards (Edwards & Chandra 1996): skin friction
SA-Edwards (Edwards & Chandra 1996): wall heat flux
$C_f \times 10^3$
$q_w$ [W/cm$^2$]
x [m] (plate leading edge at x = -1.64 m; fin apex at x = 0)
coarse grid
medium grid
fine grid
ADflow (solid)
SU2 (dotted)
Kussoy & Horstman, x = -11 cm (Edwards & Chandra's values: $C_f$ 1.0e-3, $q_w$ 1.04 W/cm$^2$)


Mach 8.2 flat plate, boundary-layer profiles at x = -0.113 m (ADflow cell centres; SU2 nodes at x = -0.107 m), medium grid
colour = model, line style = code
SA, ADflow
SA, SU2
SA-Edwards, ADflow
SA-Edwards, SU2
u/U∞
T/T∞
ν̃/ν∞
wall distance y [m]
wall distance y [m]
wall distance y [m]